

# From Memory to Testable Belief: A Survey of LLM Agents

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## Abstract

Language-model agents now act over many steps, and the failures reported from such settings are failures of state rather than of reasoning: the agent loses track of what it has learned, keeps acting on contradicted evidence, or revises what it believes without changing what it does. What an agent must hold between decisions is not a memory of what happened but a testable belief, a hypothesis that later evidence can test and revise, with three parts: a state that says what is believed and how firmly, a revision that carries it across an action, and a test that scores it against the next observation. This survey takes the belief state as its object and sorts the literature by which of the three parts a system has and where each comes from: written by the model, fixed by the designer, learned, or absent. Memory and uncertainty research gave the agent a belief state; revision mechanisms and post-training taught it to revise; the test, the part that lets the agent's own observations correct what it believes, is supplied from outside wherever it exists. We follow the three parts through construction, revision, and testing, align belief-level evaluation by construct, and identify the gap the alignment exposes: the first systems to score a written belief against the agent's own observations do so with a binary belief, in isolation, and without a framework that connects representation, testability, revision, credit, and belief-action consistency.

**AI disclosure.** The literature search, screening, and coding pipeline released with this paper is scripted and uses large-language-model agents to read and code each sampled paper under a fixed codebook (Appendix A); large-language-model assistance (Claude, Anthropic) was also used in drafting and revising the manuscript. The authors defined the framework, the inclusion criteria, and the codebook, checked the codings that support the claims in the text, and are responsible for the content.

**Table 1 | Which part of the belief each neighbouring survey supplies.**

|                    | Survey                | Organising axis                       | S | R | T | E |
|--------------------|-----------------------|---------------------------------------|---|---|---|---|
| Memory             | Zhang et al. (2024b)  | design, evaluation, use               | ■ | ■ | ■ | ■ |
|                    | Hu et al. (2025d)     | factual / experiential / working      | ■ | ■ | ■ | ■ |
|                    | Luo et al. (2026c)    | storage → reflection → experience     | ■ | ■ | ■ | ■ |
| Reward             | Wu (2025)             | reward model × stage                  | ■ | ■ | ■ | ■ |
|                    | Zheng et al. (2025a)  | data → PRM → use                      | ■ | ■ | ■ | ■ |
|                    | Zhang et al. (2025d)  | capability × application              | ■ | ■ | ■ | ■ |
|                    | Zhang (2026)          | granularity × method                  | ■ | ■ | ■ | ■ |
| Uncertainty        | Xia et al. (2025)     | four estimation avenues               | ■ | ■ | ■ | ■ |
|                    | Lin et al. (2026c)    | what / how / where                    | ■ | ■ | ■ | ■ |
| Evaluation         | Yehudai et al. (2026) | capability × domain                   | ■ | ■ | ■ | ■ |
|                    | Gao et al. (2025)     | what / when / how to evolve           | ■ | ■ | ■ | ■ |
|                    | Chen et al. (2026b)   | lifecycle × horizon carrier           | ■ | ■ | ■ | ■ |
|                    | Xu et al. (2026d)     | forecasting architecture              | ■ | ■ | ■ | ■ |
|                    | Dong et al. (2025b)   | financial function                    | ■ | ■ | ■ | ■ |
| Belief             | Cai et al. (2026)     | control-loop components               | ■ | ■ | ■ | ■ |
|                    | Tang et al. (2026c)   | decision pipeline × policy            | ■ | ■ | ■ | ■ |
| <b>This survey</b> |                       | <b>belief state · revision · test</b> | ■ | ■ | ■ | ■ |

Evaluation includes horizon, self-evolution, forecasting and finance surveys; Belief, surveys that formalise a Bayes-filter belief for one setting. S, the belief state and its persistence; R, revision, how the state is revised; T, a test that reaches the belief as a learning signal; E, belief-level evaluation. Solid, central; tint, present; grey, absent.

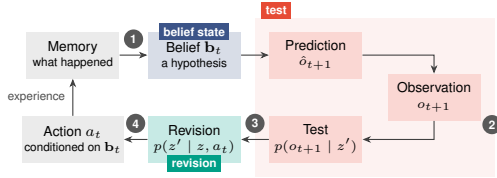
## 1 Introduction

Language-model agents now act over horizons of hours or months, and the failures reported from these settings are failures of state, not of reasoning. A model that solves a problem in one pass loses much of its accuracy when the same information arrives over several turns (Laban et al., 2025); an agent given only observations wins a fraction of the episodes won by one that reads the simulator's latent state (Alaswad et al., 2026); and in markets stronger backbones do not reliably produce stronger trading agents (Xie et al., 2024; Li et al., 2024a). What these failures share is that the task required the agent to hold something between decisions, a belief about what is true, how certain it is, and what would change it, and the agent never formed it, never tested it, or revised it without acting on the revision.

That something is a testable belief, and this is the shift the survey argues for: a memory records what has happened, whereas a belief is a hypoth-



**Fig. 1 | Taxonomy of belief-state work.** Work is split by the three parts of a testable belief, each branch by where the part comes from. The highlighted leaf holds two systems, both without uncertainty.



**Fig. 2 | From memory to testable belief.** A memory records what happened; a belief is a hypothesis that prediction, observation, and test expose to evidence. Tints mark the three parts; any evidence-based check can serve as the test, and the formulas give one realisation, the Bayes filter of Eq. 1. Numbers mark the failure modes of Section 5: 1 isolation, 2 stay, 3 update, 4 act.

esis about what is true of the world, which a prediction exposes to evidence, a test supports or falsifies, and a revision updates (Fig. 2). The test can take many forms (Section 3); this review follows three parts, a *belief state* that says what is believed and how firmly, a *revision* that carries it across an action, and a *test* that scores it against the next observation, with the Bayes filter as their principled formalisation (Åström, 1965; Kaelbling et al., 1998; Thrun et al., 2005; Friston, 2010). The line runs through memory research, not around it: a store that attaches a confidence and an update rule to its claims already has the first two parts (Zhang et al., 2024b; Luo et al., 2026c), and what makes it a belief is the third, a test that connects the hypothesis to evidence that can support or falsify it. Memory and uncertainty research gave the agent a state with a confidence (Park et al., 2023; Shinn et al.,

2023; Lin et al., 2022; Kadavath et al., 2022), and post-training taught it to revise, rewarding the response, the step, or the action (Ouyang et al., 2022; Lightman et al., 2024; Zhou et al., 2024b; Zhang, 2026). The test is still supplied from outside, by a fixed observation model, a simulator, a verifier, or a teacher.

Work on the belief itself exists, from dialogue state tracking (Lee et al., 2021) to agents that write beliefs about tasks, opponents, and markets (Kim et al., 2024; Cross et al., 2024; Yu et al., 2024) and benchmarks that score the belief rather than its outcome (Xu et al., 2026a; Deb and Krishnan, 2026; Chao et al., 2026), but it rarely cites itself (Fig. 6) and the nearest surveys organise by what is stored, stop at the action, or formalise a Bayes-filter belief for one setting at inference time only (Section 2, Table 1). This survey takes the belief state as its object and sorts the literature by which of its three parts a system has and where each comes from: written by the model, fixed by the designer, learned, or absent (Fig. 1), and follows the belief around the cycle of Fig. 2: Section 4 asks where the state lives and what form it takes; Section 5 how it is carried and revised, and where revision fails; Sections 6–8 who scores the belief against evidence, whether the score becomes a learning signal and *which object receives it*, and how evaluation measures each part. We cover 2020 to September 2026 by a scripted protocol (Appendix A); Section 10 states the open prob-

lems.

## 2 Related Work

Table 1 places this survey among sixteen neighbouring ones by which part of the belief each supplies. Memory surveys (Zhang et al., 2024b; Hu et al., 2025d; Luo et al., 2026c) supply the belief state and its persistence, the newest with a probability or revision operator on each item; what they do not ask is what evidence would retire an item, which is the test. Uncertainty surveys (Xia et al., 2025; Lin et al., 2026c) supply a confidence on the state, but attached to an output and neither carried forward nor revised. Reward and credit surveys (Wu, 2025; Zheng et al., 2025a; Zhang et al., 2025d; Zhang, 2026) supply the revision and the learning signal, following credit down to tokens, steps, and memory operations, and stop at the action. World-model and long-horizon surveys (Li et al., 2026d; Chen et al., 2026b) supply a learned revision and test whose state the agent cannot read. Evaluation, self-evolution, forecasting, and finance surveys (Yehudai et al., 2026; Gao et al., 2025; Xu et al., 2026d; Dong et al., 2025b) score the outcome and treat the belief as one component among many. Two recent surveys formalise the belief itself as a Bayes filter, for complex problem solving (Cai et al., 2026) and for proactive service agents (Tang et al., 2026c), but update it at inference time in one setting, so the test never reaches it as a learning signal. Each strand holds one part or one setting; none follows the three parts across agents into learning and evaluation, which is the question Section 3 turns into a definition.

## 3 Problem Formulation

A belief  $\mathbf{b}_t$  is a hypothesis about the environment that the agent carries between decisions. It yields a prediction, evidence  $o_{t+1}$  returns, a *test relation* supports or falsifies the hypothesis, and a revision produces  $\mathbf{b}_{t+1}$  for the next action (Fig. 2); the test may be a symbolic verifier, a consistency check, or a comparison with an external record. Partially observed decision-making gives this cycle its most principled realisation (Åström, 1965; Kaelbling et al., 1998): with hidden state  $z_t$  and history  $H_t$ , the belief is  $\mathbf{b}_t = p(z_t | H_t)$ , and a Bayes filter carries it forward as

$$\mathbf{b}_{t+1}(z') \propto p(o_{t+1} | z') \sum_z p(z' | z, a_t) \mathbf{b}_t(z) \quad (1)$$

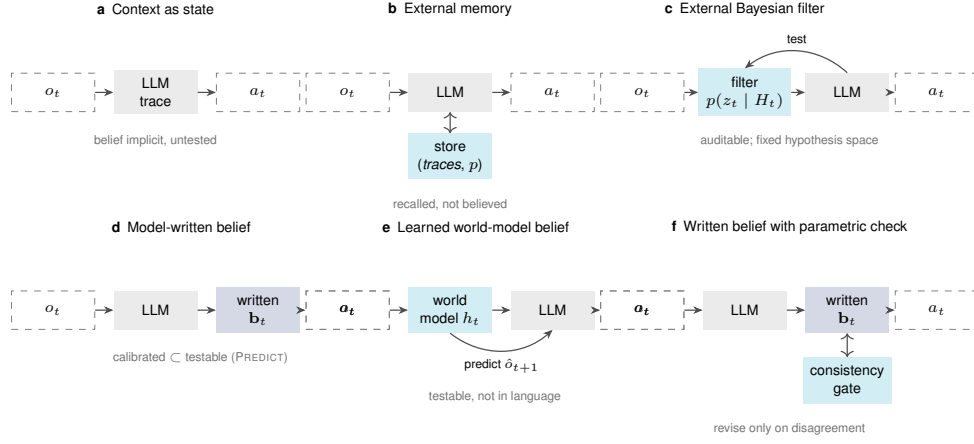
**Table 2 | Classes of the belief state and where each part comes from.**

| Locus     | Form                             | State | Uncert. | Rev. | Test                 | Belief | Example             |
|-----------|----------------------------------|-------|---------|------|----------------------|--------|---------------------|
| Context   | trace as state                   | M     | —       | M    | —                    | —      | ReAct               |
|           | written claim, calibrated        | M     | —       | M    | D (target)           | —      | Agent-BRACE, BOND   |
|           | written claim, testable          | M     | —       | M    | D (oracle)           | —      | CBM, EvoSCM         |
| Store     | trace store                      | M     | —       | D    | —                    | —      | MemGPT, A-MEM       |
|           | claims with probabilities        | M     | —       | D    | D (update rule)      | —      | Nous, BeliefMem     |
| Estimator | Bayesian filter, fixed space     | D     | —       | D    | D (closed form)      | —      | Belief-State Engine |
| Learned   | latent world model               | L     | —       | L    | L (prediction error) | —      | Dreamer, BB-WM      |
|           | written claim + parametric check | M     | —       | M    | L (check)            | —      | Hybrid-WM           |

Locus: in the context, in a store outside the model, in an external estimator, or in learned dynamics. M, written by the model; D, fixed by the designer; L, learned; —, absent (parts of Section 3). Uncert., the state carries uncertainty; Belief, meets conditions (i)–(iii) of Section 3. Solid, yes; tint, partly; grey, no.

(Thrun et al., 2005). Its three factors are the Bayesian form of the three parts the review follows: the *belief state*  $\mathbf{b}_t$ , what is believed and how firmly; *revision*, here the transition  $p(z' | z, a_t)$  that carries it across the action; and the *test*, here the likelihood  $p(o_{t+1} | z')$ ; a verifier or consistency check is a non-probabilistic test of the same kind and is coded as one. Agentic reinforcement learning casts the language-model agent as such a POMDP, with the context as history and any explicit state in a working memory the model writes (Zhang et al., 2025d; Sumers et al., 2023; Yao et al., 2023b); we write  $\mathbf{b}_t$  for whatever explicit state the span carries before the action  $a_t$ .

We call such an object a *belief state* only if it meets three conditions: (i) it is a claim about the environment carried between decisions; (ii) it carries, or admits, uncertainty; (iii) it is *testable*: it commits to a prediction that a returned observation can support or falsify. Conditions (i) and (ii) are properties of the belief state and fix the operation CONSTRUCT; condition (iii) requires a test and fixes TEST, which makes the state a hypothesis rather than a record; REVISE is the revision, which belief-revision theory constrains to change the claim without discarding the rest (Alchourrón et al., 1985; Friston, 2010). We therefore place every system by which of the three parts it has and where each comes from: written by the model, fixed by the designer, learned from data, or absent. A trace *memory* fails (i) and (ii); a store that attaches a probability and an update rule to each item meets them but scores nothing against what returns, so it fails (iii) (Luo et al., 2026c); a *world model* is itself a revisable hypothesis, a learned revision and test whose state the agent cannot read, so it meets (iii) but seldom (i) (Li et al., 2026d).



**Fig. 3 | Who maintains the belief.** Blue-grey marks a belief the model writes and can reason about; light blue marks an external estimator. Only c, d (when testable), e and f commit to a prediction that later evidence can falsify.

## 4 The Belief State: Where It Lives and Who Writes It

The belief state is placed by two attributes rather than by method family: its *locus* (in the context, in a store outside the model, in an external estimator, or in learned dynamics), which also says who writes it, and its *form* (a trace, a claim, or a distribution). The classes in Table 2 are locus–form pairs, so a system can occupy more than one, and the last column records which of conditions (i)–(iii) each class meets; Fig. 3 sketches the six designs, and Table 15 in the appendix places every system.

### 4.1 Context and Memory

The most common design has no representation at all: the reasoning trace is the state (Wei et al., 2022b; Yao et al., 2023b), the belief implicit and its uncertainty unstated.

Memory-augmented agents moved the state into a store retrieved into context (Packer et al., 2023; Park et al., 2023; Zhong et al., 2024; Xu et al., 2025d; Luo et al., 2026c; Jiang et al., 2026a). Two limits matter for belief: the stored unit is a trace rather than a claim with a probability, and an update is an operation on the store rather than on a hypothesis, so answer accuracy hides failures in the update and forget operations themselves (Hao et al., 2026). Structured stores narrow the first limit with schemas, graphs, bitemporal transitions, or AGM revision semantics (Luo et al., 2026d; Xu, 2026; Park, 2026); probabilistic stores narrow the second by attaching a number to each claim (Liao et al., 2026a; Song and Xin, 2026; Yang et al., 2026c; Zhang et al., 2026j).

### 4.2 External Bayesian Filters

A different line takes the belief out of the model. The model maps evidence to a likelihood over a fixed set of latent regimes, hypotheses, or user states, and an explicit filter maintains the posterior, whose entropy and drift can then be reported as model risk (Chattopadhyay and Halder, 2026; Dixon, 2026b; Wu et al., 2026d; Soru, 2026; Akbar et al., 2026; Azran et al., 2026; Ansell and Toney-Wails, 2026). These beliefs are validated against a closed-form oracle and auditable by construction; the price is a hypothesis space fixed in advance.

### 4.3 Model-Written Beliefs

The third line keeps the belief in context but gives it structure the model writes and reads. Dialogue state tracking wrote a slot-value belief (Lee et al., 2021), verbalised confidence attached a probability to an answer (Lin et al., 2022; Kadavath et al., 2022), agents wrote beliefs about tasks, opponents, and markets (Kim et al., 2024; Christakopoulou and Mourad, 2024; Cross et al., 2024; Yu et al., 2024), MEM1 learned to overwrite a fixed written state each turn (Zhou et al., 2025d), ReAct reflected on a goal state before each action (Kim et al., 2025), and CoEx kept a neurosymbolic belief that co-evolves with exploration (Kim and won Hwang, 2025). The newest systems add uncertainty and verification to the written object, from certainty tags and Bayesian-teacher posteriors to symbolic verification and populations of causal models (Singh et al., 2026b; Cui and Lin, 2026; Xu et al., 2026a; Zhao et al., 2026b).

Condition (iii) separates them: a written belief

**Table 3 | Four failure modes of belief revision.**

| Failure mode     | Definition                             | Part at fault                   | Measured by                   | Headline result                                             |
|------------------|----------------------------------------|---------------------------------|-------------------------------|-------------------------------------------------------------|
| Failed stay      | belief moves without evidence          | test absent: update unweighted  | BeliefShift, MEDLEY-BENCH     | consensus sensitivity from near 0 to large across 35 models |
| Failed update    | belief survives contradicting evidence | carried forward, not updated    | DeltaLogic, DToM-Track, STALE | 66.7% on a conclusion, 46.7% on whether an edit changes it  |
| Failed isolation | irrelevant input enters the belief     | belief state contaminated       | BeliefTrack, MemOps           | belief-state reward cuts failures by 71%                    |
| Failed act       | belief revised, action unchanged       | policy not conditioned on $b_t$ | STOCKTAKE, Werewolf audit     | 34–43% of correctly diagnosed weeks end in the wrong action |

For each mode: the part at fault, the benchmarks that isolate it and their headline results (Xu et al., 2026a; Chao et al., 2026; Myakala et al., 2026; Abtahi et al., 2026; Dhanda, 2026; Nguyen et al., 2026; Hao et al., 2026; Deb and Krishnan, 2026; Gao et al., 2026b).

is testable only if it names a future observation with a horizon, so that the returned observation yields a signed error. Most written beliefs are calibrated but not testable, stating confidence without naming what would falsify it; Hypothetical Minds, CBM, EvoSCM, and ReBel (Tang et al., 2026b) do.

#### 4.4 Learned World Models

The fourth line learns the dynamics and exposes the resulting belief to the policy: recurrent latent states (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2023), the language model as its own world model for tree search (Hao et al., 2023; Zhao et al., 2023b; Zhou et al., 2023a), LLM-proposed POMDPs refined to a belief-based likelihood (Six et al., 2026), and a world-model belief the policy can query (Kumar et al., 2026; Li et al., 2026d; Zuo et al., 2026). World-model beliefs are testable by construction but, unless written back, cannot be reasoned about in language. Hybrids use the learned model as a check on a written belief: a parametric consistency gate cuts the hallucinated-state rate from 0.176 to 0.035 (Song and Cai, 2026a).

**Insight 1. Structure is not testability.** Memory design has made the belief steadily more structured without making it testable, and the representations that are testable, filters and world models, cannot be reasoned about in language. The belief the field needs is written and testable at once; the first system to test one against the environment does so with a binary belief.

### 5 Revision and Its Failures

Bayesian filtering and active inference agree on how a belief should change: it predicts, evidence arrives, and the mismatch drives the update (Friston, 2010; Friston et al., 2017), routed by factor

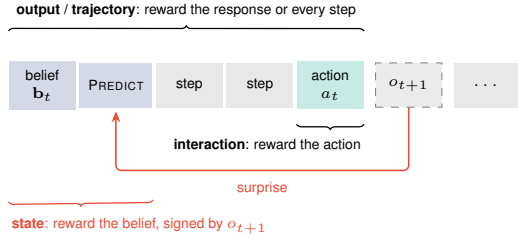
graphs to the factors that produced the prediction (Kschischang et al., 2001; Loeliger, 2004). In Equation 1 the transition carries the belief forward and the likelihood decides how far the observation moves it; the language-model literature imported the vocabulary but rarely the routing: feedback is appended to the context or folded into self-critique loops (Huang et al., 2022b; Madaan et al., 2023; Kim et al., 2023a).

#### 5.1 Four Failure Modes

Contextual Belief Management names three ways a belief can fail on new input, a *failed stay* in which the belief moves without evidence (an update no test weighs), a *failed update* in which it survives evidence against it (carried forward without an update), and a *failed isolation* in which irrelevant input contaminates it (Xu et al., 2026a); in consolidated memory a quarter or more of the agent’s own speculations resurface as user-stated facts (Verbovoy, 2026). Failed update is also the base model’s default: assistants revise correct answers under pushback (Sharma et al., 2023; Atwell et al., 2025), drift as context accumulates (Geng et al., 2025), and in context update like a filter that forgets exponentially (Zhang et al., 2025e); memory systems write the misconception in (Bensal et al., 2026). Independent studies document a fourth, the *failed act*, in which the belief is revised but the policy is not conditioned on it, so that a belief-level oracle finds the diagnosis right while the action stays wrong (Deb and Krishnan, 2026; Gao et al., 2026b). Table 3 and Fig. 2 place them.

#### 5.2 Mechanisms of Revision

The mechanisms that respond to failed update fall into three families by what triggers them. The first is triggered by *contradiction* among stored claims and resolves it with operator algebras, first-class update and forget operations, or supersession contracts (Wang, 2026b; Hao et al., 2026; Xu, 2026; Qin and Lu, 2026). The second is triggered *before commitment*, by self-audit, a parametric veto, semantic entropy, or resampling (Yuan et al., 2026a; Song and Cai, 2026a; Yan et al., 2026b; Wang et al., 2026b); these see no future observation and so cannot catch a belief that is consistent and wrong. The third is triggered by *surprise*: reward-prediction error routes writes to a fast or a slow path (Song and Xin, 2026), and prediction–observation mismatches become semantic rules (Zhang et al., 2026g); FinCon revises its invest-



**Fig. 4 | The credited object on one span of a trajectory.** State credit reaches the written belief, signed by the observation returned after the action.

**Table 4 | Learning signals by credited object and source of the test.**

| Level       | Credited object       | Test from                    | Representative                |
|-------------|-----------------------|------------------------------|-------------------------------|
| Output      | finished response     | human preference             | RLHF, DPO                     |
| Trajectory  | sampled steps         | verifier, PRM                | PRMs, GRPO, LATS              |
| Interaction | actions, tool calls   | task reward                  | ArCHer, RAGEN, HiPER          |
|             | memory operations     | task reward, verifier        | MEM1, Memory-R1, VerMem       |
| State       | written belief        | symbolic verifier            | CBM                           |
|             | claim + certainty     | calibration target           | Agent-BRACE                   |
|             | posterior over claims | Bayesian teacher             | BOND, DEL-ToM                 |
|             | state prediction      | simulator state              | VAGEN, Dyna-Think, SPA        |
|             | belief over answers   | known answer                 | T3, IGPO, InfoPO, CORE, GAVEL |
|             | written belief        | <b>own next observation</b>  | <b>ReBel</b>                  |
|             | written belief        | <b>own past observations</b> | <b>ABBEL</b>                  |

The object that receives the gradient and where the test that signs it comes from (Section 6); bold, the test comes from the agent's own observations. Table 13 in the appendix adds granularity and evidence level.

ment beliefs after each episode and sends each revision only to the agents that need it, routing the error to an agent though not yet to a claim (Yu et al., 2024).

**Insight 2. Revision is triggered, not routed.** Contradiction, self-audit, and surprise decide *when* a belief changes; none decides *which* claim the error belongs to. Until the error is routed to the claim that made the prediction, agents will keep repairing memories while beliefs stay wrong.

## 6 The Test as a Learning Signal

A learning signal is a test that has become a gradient. Fewer than three in five included papers give the belief any such signal; for the rest it is built and revised at inference time. Where existing surveys organise post-training by reward source and density (Wu, 2025; Zhang, 2026; Zheng et al., 2025a), we separate two axes that are usually run together (Fig. 4, Table 4): the *credited object*, the quantity that receives the gradient, and the *source of the test*, whatever signs it (humans, a verifier, the task reward, a teacher, a simulator, or the agent's own later observation). A method is at the *state level* if the credited object is a claim about the environment, whatever the source; the source then decides whether the sign exists outside a simulator.

### 6.1 Output and Trajectory

Supervised fine-tuning and preference optimisation credit the completed response (Ouyang et al., 2022; Wei et al., 2022a; Rafailov et al., 2023; Wang et al., 2022; Huang et al., 2022a). Trajectory-level credit began by imitating and searching over agent trajectories (Chen et al., 2023a; Zhou et al., 2023a; Putta et al., 2024; Fu et al., 2025), then credited sampled steps through PPO, GRPO, and process reward models (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024; Zheng et al., 2025a).

### 6.2 Interaction

Agentic reinforcement learning credits actions over several turns (Zhou et al., 2024b; Wang et al., 2025f; Zhou et al., 2025c; Feng et al., 2025b; Xiong et al., 2025a), and the open question is how to make that credit denser without making it wrong, by hierarchical advantages, counterfactual estimators, potential-based shaping, or verifier-structured credit (Peng et al., 2026a; Wang et al., 2026a; Diao et al., 2026; Wang et al., 2026j; Ren et al., 2026a; Liao et al., 2026b; Liu et al., 2026b; Li et al., 2026b). The nearest this level comes to the state is a group that credits the write to a memory, from a rewritten case bank to an RL-trained overwrite of a fixed written state (Zhou et al., 2025a,d; Yu et al., 2025b; Yan et al., 2025; Sun et al., 2026b; Yan et al., 2026a). These credit the *operation*, so a well-executed update to a wrong claim is still rewarded.

### 6.3 State

Until 2026 every such method took its test from outside the environment: a symbolic verifier (Xu et al., 2026a), a calibration target (Singh et al., 2026b), a Bayesian teacher (Cui and Lin, 2026; Wu et al., 2025d), the simulator's state (Wang et al., 2025d; Yu et al., 2025c; Chen et al., 2025e; Wu et al., 2025a), or a known answer (Zou et al., 2025; Wang et al., 2025a; Kong et al., 2026; Zhang et al., 2026j; Wang et al., 2026e); each needs the answer or the latent state at training time (Appendix D). Three systems now take it from the agent's own observations: ReBel rewards each predicate of a written belief that the next observation supports, with the gradient reaching the belief (Tang et al., 2026b); ABBEL grades a natural-language belief by how well the prior observation can be reconstructed from it (Lidayan et al., 2025);

**Table 5 | Evidence-acquisition decisions by what they are conditioned on.**

| Acquisition conditioned on        | Systems                                                                                                                           | Belief object                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| task-level POMDP                  | CGDP (Kausik et al., 2026), MedExAgent (Gao et al., 2026a)                                                                        | searcher, patient state                      |
| uncertainty of an explicit belief | BayesEvolve (Wu et al., 2026d), EvoSCM (Zhao et al., 2026b), Align-While-Search (Bae et al., 2025), CA-BED (Arnould et al., 2026) | hypothesis, causal, configuration posteriors |
| belief over own resources         | Bayesian-Agent (Wu et al., 2026b), REDEREF (Hosseini et al., 2026), CMI (Srivastava, 2026b), REFLEX (Wu and Lim, 2026)            | skill, delegate, memory reliability          |
| information gain as reward        | PICA (Liu et al., 2026b), TIPS (Xie et al., 2026)                                                                                 | implicit                                     |
| written testable belief           | <i>none in corpus</i>                                                                                                             | —                                            |

PaW adds a next-observation loss on the policy (Lu et al., 2026a). None carries uncertainty, and the dense form is fragile: the same reward through group-normalised advantages collapses every run, and as an auxiliary loss it is matched by shuffled targets (Wang, 2026a).

**Insight 3. Credit has reached the belief, not yet its uncertainty.** The first systems now score a written belief against the agent’s own observations, but the belief they score is binary, the dense form of that reward collapses under group normalisation, and none connects the score to revision or action. The open combination is an environment-signed test on a calibrated state.

## 7 The Test as a Choice of Evidence

A testable belief tells the agent what evidence would change it, which turns retrieval, tool use, and routing into decisions (Table 5); expected information gain needs a probabilistic test, so the systems that choose evidence by belief are those that carry one. Search and diagnosis have been framed as POMDPs whose actions include what to observe next (Kausik et al., 2026; Gao et al., 2026a), and two finance simulators show why it matters: failing agents spend 62% of their observation budget re-reading internal state (Han et al., 2026; Zhang et al., 2026c).

Agentic retrieval let the model decide when to retrieve (Gao et al., 2024); recent systems condition that decision on measured uncertainty about the task (Wu et al., 2026d; Zhao et al., 2026b; Xu and Fekri, 2026; Yan et al., 2026b; Bae et al., 2025; Arnould et al., 2026; Wang et al., 2026e), make information gain the training signal itself (Liu et al., 2026b; Xie et al., 2026), or keep a belief over the agent’s own skills, delegates, or memories (Wu et al., 2026b; Hosseini et al., 2026; Srivastava, 2026b; Wu and Lim, 2026).

|                                                             | calibration | accuracy | revision | belief-act. | outcome |
|-------------------------------------------------------------|-------------|----------|----------|-------------|---------|
| Forecast-Dojo (Ye et al., 2026)                             | ●           | ○        | ●        | ○           | ●       |
| Agent-BRACE (Singh et al., 2026b)                           | ●           | ○        | ○        | ○           | ○       |
| Belief-State Engine (Chattopadhyay and Halder, 2026)        | ●           | ○        | ○        | ●           | ○       |
| Belief at Risk (Dixon, 2026b)                               | ●           | ○        | ○        | ○           | ○       |
| FinBench (Ghosh and Devarakonda, 2026)                      | ○           | ○        | ○        | ○           | ○       |
| BeliefTrack (Xu et al., 2026a)                              | ○           | ○        | ○        | ○           | ○       |
| DeltaLogic (Dhanda, 2026)                                   | ○           | ○        | ○        | ○           | ○       |
| DTolM-Track (Nguyen et al., 2026)                           | ○           | ○        | ○        | ○           | ○       |
| BeliefShift (Myakala et al., 2026)                          | ○           | ○        | ○        | ○           | ○       |
| STALE (Chao et al., 2026)                                   | ○           | ○        | ○        | ○           | ○       |
| MemOps (Hao et al., 2026)                                   | ○           | ○        | ○        | ○           | ○       |
| STOCKTAKE (Deb and Krishnan, 2026)                          | ○           | ○        | ○        | ○           | ○       |
| Werewolf audit (Gao et al., 2026b)                          | ○           | ○        | ○        | ○           | ○       |
| HERCULEAN (Peng et al., 2026b)                              | ○           | ○        | ○        | ○           | ○       |
| FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024a) | ○           | ○        | ○        | ○           | ○       |
| BayesBench (Samanta et al., 2026)                           | ○           | ○        | ○        | ○           | ○       |
| Martingale Score (He et al., 2025)                          | ○           | ○        | ○        | ○           | ○       |
| Bayesian coherence (Imran et al., 2025)                     | ○           | ○        | ○        | ○           | ○       |
| Theory of Space (Zhang et al., 2026f)                       | ○           | ○        | ○        | ○           | ○       |
| HaluMem (Chen et al., 2025b)                                | ○           | ○        | ○        | ○           | ○       |
| Memora (Uddin et al., 2026)                                 | ○           | ○        | ○        | ○           | ○       |
| Embodied Agent Interface (Li et al., 2024b)                 | ○           | ○        | ○        | ○           | ○       |

**Fig. 5 | Belief-level constructs reported by each evaluation paper.** Only the Bayesian-coherence rows (shaded) share a reference posterior.

**Insight 4. Acquisition is a belief problem.** Agents lose by re-reading what they know rather than seeking what would change it. A testable belief names that evidence, so evidence acquisition will move from a fixed retrieval stage to a consequence of the belief.

## 8 The Test as a Metric: Evaluating Belief Apart from Outcome

A correct answer can follow from a correct belief, a wrong belief that did not matter, or a belief revised too late, and outcome metrics cannot separate the three, which is why stronger backbones do not yield stronger trading agents (Xie et al., 2024; Li et al., 2024a). Earlier agent benchmarks score task success or recall (Shridhar et al., 2020; Liu et al., 2023b; Yao et al., 2024); recent work added belief-level metrics, each defining its own; each measures one part from outside (calibration the belief state, accuracy the revision, Bayesian coherence the test, the belief–action gap whether the policy uses any). Fig. 5 aligns them by construct; Table 14 in the appendix catalogues the benchmarks.

### 8.1 Constructs

*Calibration over a trajectory* has the most statistics and the least comparability: four formulations, none computed on another paper’s system (Lin et al., 2026c; Singh et al., 2026b; Dixon, 2026b; Ye et al., 2026), and the one replicated deficit flipped sign across model versions (Joshi, 2026). *Belief accuracy and revision* are scored against a known answer, a premise edit, human annotation, or a hidden ledger (Xu et al., 2026a; Dhanda, 2026; Myakala et al., 2026; Chao et al., 2026; Srivastava, 2026a), and *memory validity* per operation (Chen et al., 2025b; Hao et al., 2026). *Bayesian coherence* compares a belief trajectory with an ideal updater (Samanta et al., 2026; He et al., 2025; Imran et al., 2025), and the *belief–action gap*, the

**Table 6 | Credit levels compared by where their test comes from.**

| Level       | Test from                     | Env. | No oracle | Belief | Act gap | Maturity |
|-------------|-------------------------------|------|-----------|--------|---------|----------|
| Output      | humans, verifiers             |      |           |        |         |          |
| Trajectory  | verifier, PRM                 |      |           |        |         |          |
| Interaction | actions                       |      |           |        |         |          |
|             | memory ops                    |      |           |        |         |          |
| State       | oracle or teacher             |      |           |        |         |          |
|             | closed world, teacher, target |      |           |        |         |          |
|             | own observation               |      |           |        |         |          |
|             | returned observation          |      |           |        |         |          |

Env., the test comes from the environment; No oracle, no external source of truth is needed; Belief, the written belief receives the gradient; Act gap, belief and action error are separated; Maturity, solid many systems, tint a few, grey none. Otherwise solid, yes; tint, partly; grey, no.

youngest construct, scores belief and action on the same episodes (Deb and Krishnan, 2026; Gao et al., 2026b); Appendix D lists every instrument.

Every construct beyond calibration needs a latent state to compare against: controlled environments provide it (Deb and Krishnan, 2026; Xu et al., 2026a; Alaswad et al., 2026; Wang et al., 2026i), long-horizon simulators know it but report belief through proxies (He et al., 2026b; Zhang et al., 2026c; Han et al., 2026), and historical replay provides neither (Peng et al., 2026b).

**Insight 5. Belief cannot be scored by outcome.** Seven constructs, more than twenty metrics, almost no cross-paper comparison, and only the fair oracle scores belief and action on the same episodes. Progress on belief will be measurable only where the latent state is known, so controlled environments, not replay, are where the next results should come from.

## 9 Discussion

Table 6 compares the credit levels by where their test comes from; three observations follow.

**From memory to testable belief: the test is scarce.** The evidence supports the survey’s central claim: agents need to move from memory, which records what happened, to belief, a hypothesis their own observations can support or falsify. Memory has become a belief state and post-training a revision, but the step from record to hypothesis is the test, and Table 4 shows the state level taking it from a source the environment does not supply in every case but two: ReBel settles a predicate belief against the next observation, ABBEL reconstructs the previous one. Both are binary and isolated. The step of Fig. 2 that makes a belief a hypothesis is no longer empty; it is thin. The cycle is broader than its Bayesian form and applies to agentic systems generally.

**Auditability and expressiveness pull apart.** Filters are validated in closed form but cannot introduce a factor they were not given, written beliefs can name a new factor but are rarely testable,

and world models are testable but mute; hybrids that write the belief and check it parametrically are the convergence point (Song and Cai, 2026a; Zhang et al., 2026g; Kumar et al., 2026).

### The belief–action gap is a training problem.

Wherever belief and action are scored on the same episodes, prompting and oracles improve the belief while action–belief consistency barely moves (Deb and Krishnan, 2026; Gao et al., 2026b); the system that moves both trains belief and policy jointly (Singh et al., 2026b). Closing the gap requires the state-level and interaction-level signals of Table 6 in one objective, not a better belief alone.

## 10 Open Problems

**A calibrated belief with an environment-signed test.** ReBel signs a written belief with the next observation, but a binary one; Agent-BRACE carries certainty, but signed by a calibration target. Joining the two without the collapse dense prediction rewards show under group normalisation is the most direct open problem.

**Latent beliefs without ground truth.** Historical data has no oracle, closed world, or annotation; filter-recovered latents and representation drift are candidate substitutes, unvalidated against each other (Dixon, 2026b; Xue, 2026); a benchmark pairing a controlled environment with a replay of the same instrument would settle whether belief-level gains transfer.

**Belief under governance.** Regulated domains add auditability, which only fixed-space filters provide; LLM-proposed spaces are the candidate bridge (Six et al., 2026; Soru, 2026; Chen et al., 2026c; Dixon, 2026c).

## 11 Conclusion

A memory records what has happened; a belief is a hypothesis about what is true, open to testing and revision by future evidence. This survey argued that agents need the second, and asked of every system which of the belief’s three parts it has and where each comes from. The answer is uneven. The field has given the agent a state, from the trace to stores, filters, and world models, and a revision, with credit made denser at every level below the belief; it has built more than twenty ways to measure belief without a shared one. The test, the step that turns a record into a hypothesis,

is still supplied from outside: the representations that are testable are not the ones the agent can reason about, the mechanisms that revise beliefs do not route the error to the claim that produced it, and the two methods that score the belief with the environment’s own returns score a binary belief and stand alone. The next step is precise: a written belief that carries uncertainty, is scored by the agent’s own matured predictions, and receives the error at the claim that made it, evaluated where the latent state is known. The taxonomy, screening record, and benchmark catalogue are released to make that step measurable.<sup>1</sup>

## Limitations

This survey covers work indexed by OpenAlex and arXiv from January 2020 to 30 September 2026, retrieved by the keyword protocol, citation-tree expansion, two targeted rounds, and the first monthly update (Appendix A); papers that describe belief maintenance without using any of our search terms may be missed. Our taxonomy assigns each method one source per part, although several systems combine sources; the codings were produced with large-language-model assistance under our codebook (Appendix A) and have not been independently replicated. Appendix A reports how each insight fares on the peer-reviewed subset alone. One directly relevant method reached us through a reader’s comment rather than the protocol; Appendix A reports the targeted search that followed. The survey reports numbers as stated by the cited papers and does not re-run any method; headline figures from preprints may change in later versions. Finance is used as the running example because delayed and partially observed evidence is the norm there; the taxonomy itself is domain-agnostic, but we have not validated it against embodied or dialogue agents in the same depth.

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<sup>1</sup><https://github.com/jiminHuang/belief-state-survey>

## Declaration of interests

Jimin Huang and Xueqing Peng are affiliated with The Fin AI, which produced several of the finance benchmarks discussed here, among them FinBen and InvestorBench; these were screened, included, and described under the same protocol as every other paper (Appendix A). One benchmark discussed in Section 8 and Table 14, Herculean (Peng et al., 2026b), is co-authored by the authors of this survey. The authors declare no other financial or personal relationships with people or organisations that could be perceived to have influenced the work reported in this paper. The funder had no role in the design of the review, the screening protocol, the analysis, or the decision to publish.

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## A Paper Collection Protocol

The search is scripted and rerunnable; the configuration (including every query string and term list), raw hits, per-stage counts, and the full-text coding record for the runs reported here (identifiers 20260915\_v1 for 2025–2026 and 20260916\_v3 for the 2020–2026 window) are released with the paper.

**Queries.** Twenty-three keyword queries were defined in the first run, four to five per survey section; a second run added eleven queries on evidence acquisition and theory-of-mind belief modelling (34 in total): representation (Section 4), revision (Section 5), learning (Section 6), evidence (Section 7), and evaluation (Section 8).

**Sources.** Source A queried OpenAlex across all venues with publication date from 1 January 2025, 50 results per query. Source B queried OpenAlex restricted to the arXiv source with publication date from 1 January 2026, 50 results per query, to improve recall of preprints. The arXiv API itself was rate-limited during the run; the OpenAlex mirror of arXiv lags by a few days.

**Screening.** Records were deduplicated by identifier and then by normalised title, so that a preprint and its journal version count once. Rule-based screening on title and abstract applied one inclusion criterion for an LLM-related term (I1), one exclusion criterion for off-domain applications (E1), and one inclusion criterion requiring at least two topic terms from a fixed list (I2). Records without an abstract of at least 200 characters were excluded. Included records were ranked by a relevance score used only for ordering the manual full-text screen, never for inclusion.

**Table 7 | Paper-collection counts for run 20260915\_v1.**

| Stage                                   | Count         |
|-----------------------------------------|---------------|
| Queries                                 | 23            |
| Source A raw hits / unique              | 1,150 / 715   |
| Source B raw hits / unique              | 1,150 / 1,008 |
| Merged unique (identifier)              | 1,685         |
| After title-level deduplication         | 1,460         |
| Excluded: no or short abstract          | 45            |
| Excluded I1: no LLM term                | 160           |
| Excluded E1: off-domain                 | 35            |
| Excluded I2: fewer than two topic terms | 357           |
| Included after rule screening           | 863           |
| of which published in 2026              | 656           |
| Full-text screened                      | 117           |
| of which included after reading         | 106           |

**Second and third runs.** The second run (34 queries, same sources and rules) returned 834 unique records, 338 after screening, of which 96 were published in 2026; only two of those had not already been read in the first run, which we take as evidence that the first run had saturated the keyword neighbourhood. The third run extended the window to 1 January 2020 with 100 results per query and returned 4,471 unique records, 3,999 after title deduplication and 2,079 after rule screening. Keyword ranking is dominated by generic high-citation papers before 2025, so the 2020–2025 foundations were enumerated as a curated list of 90 titles resolved through OpenAlex for venue and citation data (Table 16); 75 of them were read in full and 73 included.

**Citation-tree expansion.** Keyword search cannot reach papers that describe belief maintenance in other words, so the included set was expanded by snowballing over the citation graph. Each round took the papers newly included in the previous round as seeds, fetched their reference lists

(backward) and the papers citing them (forward) from the Semantic Scholar Graph API, kept candidates published from 2020, removed everything already screened, and applied the same rule-based screen as the keyword runs. Because forward citation of recent preprints is dense, the rule-included candidates were sampled for full-text reading by a fixed rule (topic score  $\geq 10$ , or the word *belief* in title or abstract, or linked to at least three seeds, or a world-model paper from 2025 with score  $\geq 7$ ), and every sampled paper was read and coded with the same fields as the keyword runs. Round one was seeded by the 179 keyword-included papers and each later round by the papers the previous round added; the full-text sample shrank from 302 to 115, 83, and 26 across four rounds, and the expansion stopped after the round whose sample fell below thirty, the stopping rule fixed in advance (Table 8).

**Table 8 | Citation-tree expansion by round.**

| Stage                         | R1     | R2     | R3    | R4    |
|-------------------------------|--------|--------|-------|-------|
| Seeds                         | 178    | 165    | 55    | 45    |
| Backward references           | 9,127  | 8,601  | 2,673 | 2,406 |
| Forward citations             | 9,999  | 9,999  | 1,403 | 441   |
| Unique candidates (2020–)     | 12,872 | 13,113 | 2,536 | 1,465 |
| Already screened              | 302    | 1,284  | 605   | 465   |
| Excluded by rules             | 8,448  | 7,401  | 1,300 | 747   |
| Included after rule screening | 4,122  | 4,428  | 631   | 253   |
| Sampled for full-text reading | 302    | 115    | 83    | 26    |
| Included after reading        | 171    | 58     | 47    | 7     |
| of which peer-reviewed        | 79     | 23     | 22    | 2     |

Forward citations are capped at 9,999 per request batch by the API.

**Fifth round, targeted.** After the four rounds above, a reader pointed us to a method the protocol had not reached (ReBel; Tang et al., 2026b), posted after most of our seeds and cited by almost no one yet. We treated it as a seed for a fifth, targeted round: the Semantic Scholar batch endpoint was rate-limited throughout, so we used ReBel’s own reference list (46 entries), an OpenAlex search on belief-consistency and belief-level supervision (85 candidates), and a Hugging Face Papers search on nine related phrases (743 candidates, 211 after the rule screen and deduplication). Full-text screening added 23 papers (3 peer-reviewed), of which four score a belief or prediction against the agent’s own observations; every paper whose signal source mattered to a claim was re-read in full. One of the 23 (PABU) had passed the rule screen in the third keyword run but was never sampled, which is the protocol’s own miss.

The additions are marked R5 in the released record and in Table 9.

**Sixth round.** A final targeted round added four papers identified by the authors and readers after the fifth round: two posted after the search cut-off, Jev-Mem and REFLEX (Jiang et al., 2026a; Wu and Lim, 2026), and two the keyword and citation searches did not reach, FLAG-TRADER and Conv-FinRe (Xiong et al., 2025a; Wang et al., 2026i). They were read in full, coded with the same fields, and are counted in the totals, the sensitivity analysis, and the master table (column R6 of Table 9).

**First monthly update.** The released record is refreshed monthly with the same OpenAlex and Hugging Face Papers queries and rule screen, restricted to the new window. The September 2026 window returned 1,051 unique records and 434 rule-screened candidates, of which 47 were read in full and 29 included (3 peer-reviewed): among them two surveys that formalise a Bayes-filter belief for one setting (Cai et al., 2026; Tang et al., 2026c) and three methods that credit a probabilistic belief against a known answer (Zhang et al., 2026j; Wang et al., 2026e; Hu et al., 2026a). They are counted throughout (column U1 of Table 9), and no insight changes direction (Table 10).

**Year distribution.** Table 9 gives the included papers by year, source, and whether peer-reviewed. The share that is peer-reviewed falls with recency, from four in five before 2025 to about one in nine for 2026, which is the ordinary lag of review rather than a property of the field.

**Table 9 | Included papers by publication year and by the run that found them (keyword search, citation-tree or targeted round, or monthly update).**

| Year  | Keyword | R1  | R2 | R3 | R4 | R5 | R6 | U1 | Total | Peer-rev. |
|-------|---------|-----|----|----|----|----|----|----|-------|-----------|
| 2020  | 1       | 3   | 1  | 0  | 0  | 0  | 0  | 0  | 5     | 4         |
| 2021  | 2       | 2   | 1  | 0  | 0  | 0  | 0  | 0  | 5     | 3         |
| 2022  | 9       | 4   | 0  | 1  | 0  | 0  | 0  | 0  | 14    | 13        |
| 2023  | 27      | 21  | 2  | 0  | 0  | 1  | 0  | 0  | 51    | 41        |
| 2024  | 16      | 20  | 9  | 4  | 1  | 1  | 0  | 0  | 51    | 40        |
| 2025  | 18      | 69  | 20 | 20 | 2  | 5  | 1  | 0  | 135   | 51        |
| 2026  | 106     | 52  | 25 | 22 | 4  | 16 | 3  | 29 | 257   | 29        |
| Total | 179     | 171 | 58 | 47 | 7  | 23 | 4  | 29 | 518   | 181       |

**Evidence level.** Of the 518 papers included after full-text reading (179 from keyword search, 283 from citation-tree expansion, 23 from the targeted fifth round, 4 from the sixth, and 29 from the first

monthly update), 181 are peer-reviewed and 337 are preprints; every number quoted in the text is the paper’s own headline figure, taken from the abstract or the reported table, and none has been reproduced by us.

**Full-text screening.** A paper passed full-text screening if it (a) concerns a language-model agent or a component used by one, and (b) constructs, revises, tests, learns, or evaluates an explicit state carried between decisions, or documents a failure of such a state; benchmarks and surveys that do (b) were kept and coded as such. Each paper read in full was recorded with the fields: who maintains the belief (source of the belief state), credited object, revision signal (source of the revision and test), whether belief-level metrics are reported, survey section, and role. The record is the source of Fig. 1, Tables 11–14, and Fig. 5. Screening and coding were carried out with large-language-model agents: each sampled paper was read and coded against these fields under the fixed codebook above, and every paper whose coding supports a claim in the text was re-read in full. The authors defined the framework, the inclusion criteria, and the codebook, and are responsible for the record; it has not been re-coded by a second annotator.

**Peer-reviewed subset.** Because 337 of the 518 papers are preprints, Table 10 recomputes the counts behind each insight box on the 181 peer-reviewed papers alone.

**Table 10 | Sensitivity of the five insights to preprint reliance: the counts each rests on, over all included papers and over the peer-reviewed subset alone.**

| # | Counts the insight rests on                       | All          | P only      |
|---|---------------------------------------------------|--------------|-------------|
| 1 | belief state written / filter / world model       | 71 / 27 / 55 | 22 / 8 / 19 |
| 2 | revision papers (§5)                              | 122          | 37          |
| 3 | credited: interaction / state / state by own obs. | 95 / 50 / 1  | 33 / 13 / 0 |
| 4 | acquisition papers (§7)                           | 66           | 18          |
| 5 | papers with a belief-level metric                 | 105          | 34          |

Every insight is a claim about a proportion or an absence, and each holds in the same direction on the peer-reviewed subset; the cell for state credit signed by the agent’s own observation holds one paper, a preprint, in both.

## B Additional Tables and Figures

## C Timeline of Learning Signals by Credited Object

## D Details for Sections 6 and 8

**State-level learning signals (Section 6.3).** Until 2026 every such method took its test from outside the environment. CBM trains with belief-state rewards from symbolic verification and cuts belief-management failures by about 71% (Xu et al., 2026a), Agent-BRACE trains belief model and policy jointly with verbalised certainty as the belief’s output (Singh et al., 2026b), and BOND distils a Bayesian teacher’s posterior into an 8B student (Cui and Lin, 2026). VAGEN rewards a written state prediction against the simulator’s state (Wang et al., 2025d), T3 truncates rollouts by a belief deviation anchored to the true hypothesis (Zou et al., 2025), IGPO and InfoPO reward a turn by the belief update it causes toward the known answer (Wang et al., 2025a; Kong et al., 2026), Dyna-Think, SPA, and RLVR-World train state prediction as a loss or verifiable reward (Yu et al., 2025c; Chen et al., 2025e; Wu et al., 2025a), DEL-ToM scores each belief-update step with a process belief model (Wu et al., 2025d), and CORE and GAVEL train a probabilistic belief against gold slots or held-out placements (Zhang et al., 2026j; Wang et al., 2026e). Each needs the answer or the latent state at training time. Three systems now take it from the agent’s own observations: ReBel rewards each predicate of a written belief that the next observation supports, with the gradient reaching the belief (Tang et al., 2026b); ABBEL grades a natural-language belief by how well the prior observation can be reconstructed from it (Lidayan et al., 2025); PaW adds a next-observation loss on the policy (Lu et al., 2026a). None carries uncertainty, and the dense form is fragile: the same reward through group-normalised advantages collapses every run, and as an auxiliary loss it is matched by shuffled targets (Wang, 2026a).

**Belief-level constructs (Section 8.1).** *Calibration over a trajectory* has the most statistics and the least comparability: checkpoint and episode calibration, posterior entropy, and proper scoring rules on time-gated forecasts express one construct four ways, none computed on another paper’s system (Lin et al., 2026c; Singh et al., 2026b; Chattopadhyay and Halder, 2026; Dixon, 2026b; Ghosh and Devarakonda, 2026; Ye et al., 2026);

**Table 11 | The belief state by who maintains it, with the revision and test each system carries.**

| Class                | Method                                               | State written by model | Append-only | Uncertainty   | Test (TEST)          | Revision (REVISE) | Domain               |
|----------------------|------------------------------------------------------|------------------------|-------------|---------------|----------------------|-------------------|----------------------|
| Context as state     | ReAct (Yao et al., 2023b)                            | ✓                      | ✓           | ✗             | ✗                    | ✗                 | general              |
| External memory      | MemGPT (Packer et al., 2023)                         | ✗                      | ✗           | ✗             | ✗                    | store update      | dialogue             |
| Probabilistic memory | BeliefMem (Liao et al., 2026a)                       | ✗                      | ✗           | prob.         | ✗                    | noisy-OR          | LoCoMo, ALFWorld     |
|                      | D-MEM (Song and Xin, 2026)                           | ✗                      | ✗           | RPE           | ✗                    | gated write       | LoCoMo               |
| External filter      | Belief-State Engine (Chattopadhyay and Halder, 2026) | ✗                      | ✗           | posterior     | ✓                    | Bayes             | Tiger, security      |
|                      | Belief at Risk (Dixon, 2026b)                        | ✗                      | ✗           | posterior     | ✓                    | Bayes             | equities             |
| Model-written        | Agent-BRACE (Singh et al., 2026b)                    | ✓                      | ✓           | ordinal       | ✗                    | RL                | ALFWorld-style       |
|                      | Linguistic forecaster (Murphy, 2026)                 | ✓                      | ✓           | numeric       | ✓                    | sequential        | ForecastBench        |
|                      | BOND (Cui and Lin, 2026)                             | ✓                      | ✓           | posterior tag | ✗                    | distilled         | negotiation          |
|                      | CBM (Xu et al., 2026a)                               | ✓                      | ✓           | ✗             | ✓ (symbolic)         | RL                | closed world         |
|                      | EvoSCM (Zhao et al., 2026b)                          | ✓                      | ✓           | population    | ✓ (interventions)    | rule distillation | science              |
|                      | ReBel (Tang et al., 2026b)                           | ✓                      | ✗           | ✗             | ✓ (next observation) | RL                | ALFWorld, WebShop    |
|                      | ABBEL (Lidayan et al., 2025)                         | ✓                      | ✗           | ✗             | reconstruction       | RL                | text games, ColBench |
|                      | World model                                          | ✗                      | ✗           | latent        | ✓                    | learned           | general              |
|                      | BB-WM (Kumar et al., 2026)                           | ✗                      | ✗           | latent        | ✓                    | learned           | general              |

TEST: the belief commits to a testable prediction.

**Table 12 | Belief revision methods and benchmarks by failure mode addressed, revision trigger, and where the signal is projected.**

| Method                             | Failed stay | Failed update | Failed isolation | Failed act | Trigger             | Signal projected to |
|------------------------------------|-------------|---------------|------------------|------------|---------------------|---------------------|
| Reflexion (Shinn et al., 2023)     |             | ✓             |                  |            | task failure        | whole context       |
| BeliefShift (Myakala et al., 2026) | ✓           | ✓             |                  |            | user evidence       | (benchmark)         |
| STALE (Chao et al., 2026)          |             | ✓             |                  | ✓          | implicit conflict   | (benchmark)         |
| CBM (Xu et al., 2026a)             | ✓           | ✓             | ✓                |            | symbolic check      | belief state        |
| TOKI (Wang, 2026b)                 |             | ✓             |                  |            | contradiction       | memory rows         |
| MemOps (Hao et al., 2026)          |             | ✓             | ✓                |            | lifecycle op        | (benchmark)         |
| D-MEM (Song and Xin, 2026)         |             | ✓             | ✓                |            | reward pred. error  | memory path         |
| STOCKTAKE (Deb and Krishnan, 2026) |             | ✓             |                  | ✓          | oracle mismatch     | (benchmark)         |
| Agent-BRACE (Singh et al., 2026b)  |             | ✓             |                  | ✓          | certainty           | belief model        |
| Werewolf audit (Gao et al., 2026b) |             | ✓             |                  | ✓          | external belief log | (benchmark)         |

one replicated deficit flipped sign across model versions (Joshi, 2026). *Belief accuracy and revision* are measured against a known answer in a closed world, after a minimal premise edit, and across recall, inference, and change detection (Xu et al., 2026a; Dhanda, 2026; Nguyen et al., 2026), against human annotation or a hidden ledger (Myakala et al., 2026; Chao et al., 2026; Hao et al., 2026; Yang et al., 2026b; Srivastava, 2026a), a probed spatial map or a screenplay’s state (Zhang et al., 2026f; Tian et al., 2026), and theory-of-mind benchmarks (Kim et al., 2023b; Bawatneh et al., 2026; Wang et al., 2026c). *Memory validity* is scored per operation or by penalising invalidated memories (Chen et al., 2025b; Wu et al., 2024; Liu et al., 2026a; Uddin et al., 2026). *Bayesian coherence* compares a belief trajectory with an ideal updater: total-variation distance to the exact posterior (Samanta et al., 2026), a martingale test on successive beliefs (He et al., 2025), and the correlation between the model’s update and the log-odds of the evidence (Imran et al., 2025; Yuan et al., 2025b). The *belief–action gap* is the youngest construct: STOCKTAKE scores the agent against an exact Bayes filter on the same observations (Deb and Krishnan, 2026), the Werewolf audit reports action–belief consistency (Gao et al., 2026b), and SimpleToM separates inferring

a belief from applying it (Gu et al., 2024).

**Table 13 | Learning signals by credited object, with the source of the test that signs them.**

| Level       | Representative methods                                                                                           | Test from                           | Granularity                     | Signed by returned $\sigma_{t+1}$ | Delayed evidence   | Partial obs. |
|-------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------|-----------------------------------|--------------------|--------------|
| Output      | SFT, RLHF, DPO (Ouyang et al., 2022; Rafailov et al., 2023)                                                      | demos, preferences                  | response                        | $\times$                          | $\times$           | $\times$     |
| Trajectory  | PPO, GRPO, PRMs (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024)                                | verifier, PRM                       | step / rollout                  | $\times$                          | $\times$           | $\times$     |
| Interaction | HiPER, G <sup>2</sup> PO, ECPO, IAPO (Peng et al., 2026a; Wang et al., 2026; Li et al., 2026; Ren et al., 2026a) | environment                         | turn / action                   | $\times$                          | partial            | $\times$     |
| State       | Memory-R2, TAMTRL, EvoHarness-RL (Yan et al., 2026a; Wang et al., 2026d; Ning et al., 2026)                      | environment, teacher                | memory operation                | $\times$                          | partial            | $\checkmark$ |
|             | CBM (Xu et al., 2026a)                                                                                           | symbolic belief reward              | belief state                    |                                   | closed world       | $\checkmark$ |
|             | Agent-BRACE (Singh et al., 2026b)                                                                                | joint RL, calibration               | claim + certainty               |                                   | calibration target | $\checkmark$ |
|             | BOND (Cui and Lin, 2026)                                                                                         | Bayesian teacher                    | posterior tag                   |                                   | teacher            | $\times$     |
|             | CORE (Zhang et al., 2026j)                                                                                       | gold slot values                    | probabilistic slot belief       |                                   | known answer       | $\times$     |
|             | GAVEL (Wang et al., 2026e)                                                                                       | held-out placements                 | posterior over object locations |                                   | known answer       | $\times$     |
|             | ReBel (Tang et al., 2026b)                                                                                       | own next observation                | predicate belief                |                                   |                    | $\checkmark$ |
|             | ABBEL (Lidayan et al., 2025)                                                                                     | reconstruction of prior observation | natural-language belief         | $\times$ (backward)               |                    | $\checkmark$ |
|             | PaW (Lu et al., 2026a)                                                                                           | own next observation                | prediction on the policy        | $\checkmark$                      | $\times$           | $\checkmark$ |
|             |                                                                                                                  |                                     |                                 |                                   |                    |              |

**Table 14 | Benchmarks that exercise belief maintenance, by whether the evaluator can observe the latent state and what belief-level quantity is reported.**

| Benchmark                                                               | Domain                  | Horizon           | Latent state            | Observable to evaluator     | Delayed evidence | Belief-level quantity reported                           |
|-------------------------------------------------------------------------|-------------------------|-------------------|-------------------------|-----------------------------|------------------|----------------------------------------------------------|
| ALFWorld (Shridhar et al., 2020)                                        | text + embodied         | episodic          | household state         | $\checkmark$                | partial          | none (task success)                                      |
| PlanBench (Valmeekam et al., 2022)                                      | classical planning      | plan-length       | PDDL state              | $\checkmark$                | $\times$         | plan verification, replanning                            |
| AgentBench (Liu et al., 2023b)                                          | 8 environments          | episodic          | environment state       | $\checkmark$                | partial          | none (task success)                                      |
| SmartPlay (Wu et al., 2023)                                             | 6 games                 | episodic          | game state              | $\checkmark$                | partial          | none (9 capability scores)                               |
| T4D / FaR (Zhou et al., 2023b)                                          | theory of mind          | single-turn       | others' beliefs         | $\checkmark$ (annotated)    | $\times$         | belief tracking vs. action selection                     |
| False-belief tasks (Kosinski, 2023)                                     | theory of mind          | single-turn       | others' beliefs         | $\checkmark$ (annotated)    | $\times$         | false-belief accuracy                                    |
| LoCoMo (Maharana et al., 2024)                                          | long conversations      | 35 sessions       | conversation facts      | annotated                   | $\checkmark$     | temporal QA (proxy)                                      |
| $\tau$ -bench / $\tau^2$ -bench (Yao et al., 2024; Barres et al., 2025) | tool-user interaction   | multi-turn        | database state          | $\checkmark$                | $\checkmark$     | pass*k consistency; reasoning vs. communication errors   |
| MemoryAgentBench (Hu et al., 2025c)                                     | incremental multi-turn  | long-horizon      | memory contents         | annotated                   | $\checkmark$     | retrieval, test-time learning, conflict resolution       |
| BeliefTrack (Xu et al., 2026a)                                          | closed-world reasoning  | multi-turn        | finite belief space     | $\checkmark$ (symbolic)     | $\checkmark$     | turn-level belief accuracy; failed stay/update/isolation |
| DeltaLogic (Dhanda, 2026)                                               | logical reasoning       | 2 steps           | premise set             | $\checkmark$                | $\checkmark$     | revision accuracy vs. initial accuracy                   |
| DTOM-Track (Nguyen et al., 2026)                                        | dialogue                | multi-turn        | interlocutor belief     | $\checkmark$ (annotated)    | $\checkmark$     | prior-belief recall; change detection                    |
| BeliefShift (Myakala et al., 2026)                                      | user modelling          | multi-session     | user opinion            | annotated                   | $\checkmark$     | revision accuracy; drift coherence; evidence sensitivity |
| OmniToM (Bawatneh et al., 2026)                                         | theory of mind          | narrative         | actors' beliefs         | $\checkmark$ (annotated)    | $\times$         | belief-proposition extraction; schema labels             |
| MINDGAMES (Wang et al., 2026c)                                          | social games            | repeated games    | hidden roles, opponents | $\checkmark$                | $\checkmark$     | TrueSkill; turn-level logs                               |
| MEDLEY-BENCH (Abtahi et al., 2026)                                      | social pressure         | 2 rounds          | —                       | $\times$                    | $\times$         | sensitivity to manipulated consensus                     |
| STALE (Chao et al., 2026)                                               | memory validity         | multi-turn        | fact validity           | annotated                   | $\checkmark$     | state resolution; premise resistance; policy adaptation  |
| MemOps (Hao et al., 2026)                                               | long conversations      | multi-session     | memory lifecycle        | annotated                   | $\checkmark$     | operation-level probes                                   |
| GroupMemBench (Yang et al., 2026b)                                      | multi-party dialogue    | multi-session     | per-speaker knowledge   | annotated                   | $\checkmark$     | information-update accuracy                              |
| MemGym (Xu et al., 2026c)                                               | mixed agent tasks       | long-horizon      | —                       | $\times$                    | partial          | memory-isolated score delta                              |
| STOCKTAKE (Deb and Krishnan, 2026)                                      | inventory control       | weekly, long      | demand factors          | $\checkmark$ (Bayes filter) | $\checkmark$     | detection lag; skill vs. oracle; knowing-doing rate      |
| Flux (Alaswad et al., 2026)                                             | text game               | long-horizon      | simulator state         | $\checkmark$                | $\checkmark$     | state-tracking error modes                               |
| YC-Bench (He et al., 2026b)                                             | startup simulation      | hundreds of turns | market, clients         | privileged outcome          | $\checkmark$     | scratchpad usage (proxy)                                 |
| RetailBench (Zhang et al., 2026c)                                       | retail POMDP            | 180 days          | demand, supply          | privileged oracle           | $\checkmark$     | evidence-acquisition behaviour (proxy)                   |
| EnterpriseArena (Han et al., 2026)                                      | lending                 | 132 months        | borrower pool           | privileged outcome          | $\checkmark$     | observation-budget allocation (proxy)                    |
| Business Arena (Pan et al., 2026a)                                      | marketplace             | long-horizon      | market                  | privileged outcome          | $\checkmark$     | action-level attribution                                 |
| Werewolf audit (Gao et al., 2026b)                                      | social deduction        | 9-player game     | roles                   | $\checkmark$                | $\checkmark$     | action-belief consistency                                |
| CAGE-2 study (Bogdanov et al., 2026)                                    | cyber defence           | episodic          | network state           | $\checkmark$                | partial          | state-abstraction return gain                            |
| MD5 state tracking (Pai and Xian, 2026)                                 | tool chains             | 196 calls         | registers               | $\checkmark$                | $\times$         | exact-state success                                      |
| FinBench (time-gated) (Ghosh and Devarakonda, 2026)                     | asset forecasting       | daily             | returns                 | realised                    | $\checkmark$     | Brier; interval scores                                   |
| TradeArena (Xue, 2026)                                                  | trading                 | multi-week        | regimes                 | realised                    | $\checkmark$     | representation drift (early warning)                     |
| HERCULEAN (Peng et al., 2026b)                                          | finance workflows       | multi-step        | market, books           | partial                     | $\checkmark$     | state-tracking sub-score                                 |
| FinalityBench (Sharma, 2026)                                            | financial operations    | multi-step        | canonical event log     | $\checkmark$                | $\checkmark$     | none (effect-level loss)                                 |
| Forecast-Dojo (Ye et al., 2026)                                         | event forecasting       | dated timeline    | event outcomes          | $\checkmark$ (resolution)   | $\checkmark$     | per-step Brier, log loss                                 |
| EnterpriseBench (Yang et al., 2026d)                                    | enterprise decisions    | interactive       | hidden information      | privileged                  | $\checkmark$     | none (task level)                                        |
| Ledger memory benchmark (Srivastava, 2026a)                             | memory                  | multi-session     | principals' beliefs     | $\checkmark$ (ledger)       | $\checkmark$     | expected belief state per principal                      |
| Werewolf accusations (Yang et al., 2026f)                               | social deduction        | single message    | player roles            | $\checkmark$                | $\times$         | prior suspicion and shift per accusation                 |
| Regent Chess (Joshi, 2026)                                              | board game              | game              | hidden royal piece      | $\checkmark$                | $\checkmark$     | action-time Brier                                        |
| FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024a)             | finance tasks / trading | task / episodic   | —                       | $\times$                    | partial          | none (outcome only)                                      |

"Annotated" means human labels stand in for the latent state; "privileged" means the simulator knows it but the benchmark reports only outcomes.

**Table 15 | Systems with an explicit belief mechanism, by year.**

| System                                               | Year | State kept by | Credited         | Revision trigger                 | Metric       |
|------------------------------------------------------|------|---------------|------------------|----------------------------------|--------------|
| Cyclic subtask graphs (Gharzaddine and Saab, 2026)   | 2026 | written       | none             | state-analysis-and-routing...    |              |
| CHAL (Giovannelli and Kent, 2026)                    | 2026 | written       | none             | gradient-informed revision on... | o            |
| Pinductor (Six et al., 2026)                         | 2026 | world model   | world-model loss | belief-based likelihood score... | o            |
| Belief Engine stance... (Yang et al., 2026c)         | 2026 | prob. memory  | none             | log-odds update with evidence... | $\checkmark$ |
| HiPF (Diao et al., 2026)                             | 2026 | written       | trajectory       | subgoal completion +...          |              |
| ProFrag (Ma et al., 2026b)                           | 2026 | world model   | trajectory       | post-episode graph refinement... | o            |
| POMDP validation... (Dixon, 2026c)                   | 2026 | ext. filter   | none             | Bayesian                         | $\checkmark$ |
| Internalizing the Future (Zhang et al., 2026h)       | 2026 | written       | trajectory       | foresight-conditioned RL...      | o            |
| Hybrid-WM (Song and Cai, 2026a)                      | 2026 | world model   | world-model loss | consistency gate                 | $\checkmark$ |
| Governance-aware POMDP... (Dixon, 2026a)             | 2026 | ext. filter   | none             | Bayesian update on evidence...   | o            |
| BayesEvolve (Wu et al., 2026d)                       | 2026 | ext. filter   | none             | posterior refit after each...    | $\checkmark$ |
| WorldEvolver (Zhang et al., 2026g)                   | 2026 | world model   | world-model loss | prediction-observation...        | o            |
| WM-SAR (Song and Cai, 2026b)                         | 2026 | ext. filter   | trajectory       | spectral error amplification     |              |
| BCO (Chen et al., 2026c)                             | 2026 | written       | none             | evaluation of new candidate...   | $\checkmark$ |
| EvosCM causal belief... (Zhao et al., 2026b)         | 2026 | written       | none             | prediction-observation...        | $\checkmark$ |
| CAPTURE (Hossain et al., 2026)                       | 2026 | world model   | world-model loss | uncertainty-triggered...         | $\checkmark$ |
| Semantic Bayesian World... (Soru, 2026)              | 2026 | ext. filter   | none             | Bayesian conditioning on...      |              |
| Belief-State Engine (Chattopadhyay and Halder, 2026) | 2026 | ext. filter   | none (prompting) | Bayesian posterior               | $\checkmark$ |
| ReBel (Tang et al., 2026b)                           | 2026 | written       | state            | next observation (consistency)   | o            |
| ABBEL (Lidayan et al., 2025)                         | 2026 | written       | state            | reconstruction of prior obs.     | o            |
| PABU (Uyang et al., 2026c)                           | 2026 | written       | state            | teacher progress labels          | o            |
| MARBO (Hwang et al., 2026)                           | 2026 | written       | state            | ground-truth roles               | o            |
| CORE (Zhang et al., 2026i)                           | 2026 | prob. memory  | state            | gated slot revision              | $\checkmark$ |
| GAVEL (Wang et al., 2026e)                           | 2026 | ext. filter   | state            | online Bayesian update           | $\checkmark$ |
| BCA (Akbar et al., 2026)                             | 2026 | ext. filter   | none             | Beta update on appraised...      | $\checkmark$ |
| MoM / P-Mem (Qin and Lu, 2026)                       | 2026 | written       | none             | typed write-time operations      | $\checkmark$ |

State kept by: who maintains the belief (Section 4); Credited: the object that receives the learning signal under the criterion of Section 6, *none* if the belief is not trained; Metric:  $\checkmark$  reports a belief-level quantity, o a proxy.

**Table 16 | Every paper included after full-text reading, ordered by year and first author.**

| System                                                       | Year | State kept by | Credited    | Revision trigger                             | Metric |
|--------------------------------------------------------------|------|---------------|-------------|----------------------------------------------|--------|
| GATA belief graphs (Adhikari et al., 2020)                   | 2020 | world model   | state       | graph updater infers and updates latent. . . | ○      |
| Text games with common sense (Dambekodi et al., 2020)        | 2020 | ext. memory   | trajectory  | commonsense inference (COMET/BERT) fill. . . | ○      |
| RAG (Lewis et al., 2020)                                     | 2020 | ext. memory   | output      | retrieved passages condition generation. . . |        |
| ToM and decision science typology (Rusch et al., 2020)       | 2020 | –             | –           | none                                         |        |
| ALFWorld (Shridhar et al., 2020)                             | 2020 | –             | output      | n/a (environment)                            |        |
| Decision Transformer (Chen et al., 2021)                     | 2021 | context       | interaction | return-to-go conditioning over past sta. . . |        |
| Do LMs have beliefs (Hase et al., 2021)                      | 2021 | –             | state       | belief-update objective (SLAG) applied. . .  | ✓      |
| BeliefBank (Kassner et al., 2021)                            | 2021 | ext. memory   | –           | weighted MaxSAT revises beliefs that cl. . . | ✓      |
| Schema-driven DST (Lee et al., 2021)                         | 2021 | written       | state       | dialogue history plus schema descriptio. . . | ✓      |
| WebGPT (Nakano et al., 2021)                                 | 2021 | context       | trajectory  | none (browsing history in context; rewa. . . |        |
| SayCan (Ahn et al., 2022)                                    | 2022 | –             | interaction | affordance value functions score skill. . .  |        |
| LMs as Agent Models (Andreas, 2022)                          | 2022 | –             | –           | n/a (position paper)                         |        |
| Towards Teachable Reasoning Syste. . . (Mishra et al., 2022) | 2022 | ext. memory   | –           | user corrections of erroneous model bel. . . | ○      |
| Inner Monologue (Huang et al., 2022b)                        | 2022 | context       | interaction | success detection, scene description, h. . . |        |
| LLMs Can Self-Improve (Huang et al., 2022a)                  | 2022 | –             | output      | self-consistency majority vote as pseud. . . |        |
| LMs (Mostly) Know (Kadavath et al., 2022)                    | 2022 | written       | output      | self-evaluation P(True) and P(IK) of ow. . . | ✓      |
| Verbalized Uncertainty (Lin et al., 2022)                    | 2022 | written       | output      | fine-tuning on calibrated verbal confid. . . | ✓      |
| HELM (Paischer et al., 2022)                                 | 2022 | ext. memory   | trajectory  | frozen LM compresses observation histor. . . |        |
| ILQL (Snell et al., 2022)                                    | 2022 | context       | trajectory  | none (offline returns from logged dialo. . . |        |
| PlanBench (Valmeekam et al., 2022)                           | 2022 | –             | output      | n/a (benchmark; IPC planning domains)        |        |
| Self-Consistency (Wang et al., 2022)                         | 2022 | –             | output      | marginalisation over sampled reasoning. . .  |        |
| Chain-of-Thought (Wei et al., 2022b)                         | 2022 | context       | output      | none (single forward pass; intermediate. . . |        |
| WebShop (Yao et al., 2022a)                                  | 2022 | context       | –           | none (environment; partial observabilit. . . |        |
| ReAct (Yao et al., 2022b)                                    | 2022 | context       | –           | reasoning traces used to induce, track. . .  |        |
| Self-RAG (Asai et al., 2023)                                 | 2023 | context       | output      | self-assessed retrieval need (reflectio. . . | ○      |
| NLMs as epistemic agents (Betz et al., 2023)                 | 2023 | –             | –           | novel evidence item propagated through. . .  | ✓      |
| Self-Debug (Chen et al., 2023b)                              | 2023 | context       | output      | execution feedback + self-generated cod. . . |        |
| FireAct (Chen et al., 2023a)                                 | 2023 | context       | trajectory  | supervised fine-tuning on GPT-4 ReAct/C. . . |        |
| DreamerV3 (Hafner et al., 2023)                              | 2023 | world model   | state       | RSSM latent posterior updated from obse. . . |        |
| RAP (Hao et al., 2023)                                       | 2023 | world model   | trajectory  | LLM-as-world-model predicted next state. . . |        |
| HI-TOM (He et al., 2023)                                     | 2023 | context       | –           | none (story-based false-belief question. . . | ✓      |
| LAW (Hu and Shu, 2023)                                       | 2023 | world model   | –           | position paper: world and agent models. . .  |        |
| RCI (Kim et al., 2023a)                                      | 2023 | context       | interaction | recursive self-criticism of proposed ac. . . |        |
| FANToM (Kim et al., 2023b)                                   | 2023 | context       | –           | none (evaluation)                            | ✓      |
| LLM ToM (Kosinski) (Kosinski, 2023)                          | 2023 | –             | output      | n/a (probing)                                | ✓      |
| Semantic Entropy (Kuhn et al., 2023)                         | 2023 | –             | output      | entropy over meaning-clustered samples       | ✓      |
| RecallM (Kynoch et al., 2023)                                | 2023 | ext. memory   | memory op   | new knowledge statement triggers concep. . . | ✓      |
| TradingGPT (Li et al., 2023b)                                | 2023 | ext. memory   | interaction | layered memory decay + inter-agent deba. . . |        |
| ToM Multi-Agent Collab (Li et al., 2023a)                    | 2023 | written       | –           | prompted explicit belief-state update e. . . | ✓      |
| Let’s Verify Step by Step (Lightman et al., 2024)            | 2023 | –             | output      | per-step human process labels (PRM800K)      |        |
| AgentBench (Liu et al., 2023b)                               | 2023 | –             | output      | n/a (benchmark)                              |        |
| LLM+P (Liu et al., 2023a)                                    | 2023 | –             | output      | n/a (translate to PDDL, symbolic planne. . . |        |
| RAFA (Liu et al., 2023c)                                     | 2023 | context       | –           | switching condition: replan when predic. . . | ○      |
| Self-Refine (Madaan et al., 2023)                            | 2023 | context       | output      | self-generated natural-language feedbac. . . |        |
| RET-LLM (Modarressi et al., 2023)                            | 2023 | ext. memory   | –           | LLM emits memory-write of extracted tri. . . |        |
| DECKARD (Nottingham et al., 2023)                            | 2023 | world model   | WM loss     | LLM-hypothesized abstract world model v. . . | ○      |
| MemGPT (Packer et al., 2023)                                 | 2023 | ext. memory   | output      | self-directed memory paging via functio. . . |        |
| Generative Agents (Park et al., 2023)                        | 2023 | ext. memory   | interaction | observation stream; periodic reflection. . . |        |
| DPO (Rafailov et al., 2023)                                  | 2023 | –             | output      | none                                         |        |
| SymbolicToM (Sclar et al., 2023)                             | 2023 | ext. filter   | –           | symbolic graph update on each narrated. . .  | ✓      |
| Clever Hans N-ToM (Shapira et al., 2023)                     | 2023 | context       | –           | none (evaluation)                            | ○      |
| Sycophancy (Sharma et al., 2023)                             | 2023 | context       | output      | user pushback / stated user view (socia. . . | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 17 | Master table, continued (2/10).**

| System                                                     | Year | State kept by | Credited    | Revision trigger                           | Metric |
|------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| Reflexion (Shinn et al., 2023)                             | 2023 | ext. memory   | trajectory  | verbal self-reflection after episode fa... |        |
| CoALA (Sumers et al., 2023)                                | 2023 | ext. memory   | –           | conceptual: learning actions update mem... |        |
| Interactive planning POMDP (Sun et al., 2023)              | 2023 | context       | –           | none                                       |        |
| LLaRP (Szot et al., 2023)                                  | 2023 | context       | trajectory  | none (egocentric observations in contex... |        |
| Voyager (Wang et al., 2023a)                               | 2023 | ext. memory   | trajectory  | environment execution feedback, errors,... |        |
| DEPS (Wang et al., 2023c)                                  | 2023 | context       | trajectory  | descriptor of execution outcome; self-e... |        |
| Math-Shepherd (Wang et al., 2023b)                         | 2023 | –             | output      | automatic per-step labels via MC comple... |        |
| SimToM (Wilf et al., 2023)                                 | 2023 | context       | –           | perspective-taking filter: context rest... | ○      |
| SmartPlay (Wu et al., 2023)                                | 2023 | –             | output      | n/a (benchmark; game environments)         |        |
| Tree of Thoughts (Yao et al., 2023a)                       | 2023 | context       | trajectory  | self-evaluation of intermediate thought... |        |
| FinMem (Yu et al., 2023)                                   | 2023 | ext. memory   | interaction | layered memory with recency/relevancy/i... |        |
| AgentTuning (Zeng et al., 2023)                            | 2023 | context       | trajectory  | hybrid instruction tuning on AgentInstr... |        |
| CoELA (Zhang et al., 2023b)                                | 2023 | ext. memory   | trajectory  | semantic memory of world/agents/tasks u... |        |
| ProAgent (Zhang et al., 2023a)                             | 2023 | written       | –           | teammate intention inferred from observ... | ○      |
| LLM-MCTS (Zhao et al., 2023b)                              | 2023 | world model   | trajectory  | LLM-sampled world state belief (object...  |        |
| ExpeL (Zhao et al., 2023a)                                 | 2023 | ext. memory   | trajectory  | cross-task insight extraction from succ... |        |
| Synapse (Zheng et al., 2023)                               | 2023 | ext. memory   | –           | none (exemplar retrieval by similarity)    |        |
| MemoryBank (Zhong et al., 2023a)                           | 2023 | ext. memory   | output      | Ebbinghaus-style forgetting with time d... |        |
| MQuAKE / MeLLO (Zhong et al., 2023b)                       | 2023 | ext. memory   | –           | edited fact stored externally; iterativ... | ✓      |
| FaR / T4D (Zhou et al., 2023b)                             | 2023 | context       | interaction | foresee-and-reflect structured promptin... |        |
| RecurrentGPT (Zhou et al., 2023d)                          | 2023 | written       | output      | LLM rewrites its own natural-language s... |        |
| LATS (Zhou et al., 2023a)                                  | 2023 | ext. memory   | trajectory  | MCTS backup of LM-valued rewards + self... |        |
| WebArena (Zhou et al., 2023c)                              | 2023 | context       | –           | none                                       | ○      |
| DigiRL (Bai et al., 2024)                                  | 2024 | context       | interaction | doubly-robust step-level advantage mixi... |        |
| Latent state estimation for UI ag... (Bishop et al., 2024) | 2024 | written       | –           | none                                       |        |
| WMA Web Agent (Chae et al., 2024)                          | 2024 | world model   | WM loss     | predicted next-observation as free-form... | ○      |
| NegotiationToM (Chan et al., 2024)                         | 2024 | –             | –           | none                                       | ✓      |
| ToMBench (Chen et al., 2024b)                              | 2024 | –             | –           | none                                       | ○      |
| Agent-FLAN (Chen et al., 2024a)                            | 2024 | context       | trajectory  | none (SFT on decomposed agent corpora;...  | ○      |
| Talker-Reasoner (Christakopoulou and Mourad, 2024)         | 2024 | written       | state       | Reasoner (System 2) writes/updates user... |        |
| Hypothetical Minds (Cross et al., 2024)                    | 2024 | written       | state       | hypotheses about other agents’ strategi... |        |
| GIF-MCTS code world models (Dainese et al., 2024)          | 2024 | world model   | WM loss     | unit-test and trajectory feedback drive... | ○      |
| InferAct (Fang et al., 2024)                               | 2024 | written       | interaction | ToM-style belief inference over the age... | ○      |
| SimpleToM (Gu et al., 2024)                                | 2024 | –             | –           | none                                       | ✓      |
| Rational belief revision in model... (Hase et al., 2024)   | 2024 | –             | –           | model edits evaluated against an ideali... | ✓      |
| From Words to Actions (He et al., 2024)                    | 2024 | context       | –           | in-context Bayesian aggregated imitatio... |        |
| Coherence norms (Hofweber et al., 2024)                    | 2024 | –             | –           | coherence norms on credence (next-token... |        |
| TimeToM (Hou et al., 2024)                                 | 2024 | written       | –           | temporal ordering of events; belief com... | ○      |
| HiAgent (Hu et al., 2024)                                  | 2024 | written       | –           | subgoal completion judged by the LLM tr... | ○      |
| MMToM-QA (Jin et al., 2024)                                | 2024 | prob. memory  | –           | Bayesian inverse planning updates poste... | ✓      |
| PercepToM (Jung et al., 2024)                              | 2024 | written       | –           | perception inference feeds a perception... | ✓      |
| QuBE (Kim et al., 2024)                                    | 2024 | written       | state       | question-answering over observations to... |        |
| VisualWebArena (Koh et al., 2024)                          | 2024 | –             | output      | n/a (benchmark)                            |        |
| SCoRe (Kumar et al., 2024)                                 | 2024 | context       | interaction | multi-turn online RL with a reward bonu... |        |
| ReadAgent (Lee et al., 2024)                               | 2024 | ext. memory   | –           | agent decides episode boundaries, compr... |        |
| Embodied Agent Interface (Li et al., 2024b)                | 2024 | world model   | –           | none (evaluation only)                     | ○      |
| MindForge (Lic et al., 2024)                               | 2024 | written       | –           | new percepts and inter-agent messages u... |        |
| LoCoMo (Maharana et al., 2024)                             | 2024 | –             | output      | n/a (benchmark; QA/event summarization...  |        |
| Agent Q (Putta et al., 2024)                               | 2024 | ext. memory   | trajectory  | MCTS with AI self-critique feedback + o... |        |
| WebRL (Qi et al., 2024)                                    | 2024 | –             | output      | outcome-supervised reward model; self-g... |        |
| r to Q* (Rafailov et al., 2024)                            | 2024 | –             | output      | none (token-level implicit reward)         |        |
| Belief state geometry (Shai et al., 2024)                  | 2024 | context       | –           | implicit Bayesian belief updating over...  | ✓      |

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 18 | Master table, continued (3/10).

| System                                                  | Year | State kept by | Credited    | Revision trigger                           | Metric |
|---------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| MuMA-ToM (Shi et al., 2024)                             | 2024 | world model   | –           | Bayesian inverse multi-agent planning o... | ✓      |
| FortUne Dial (Sicilia et al., 2024)                     | 2024 | context       | output      | fine-tuning (supervised and off-policy...  | ✓      |
| Common-ToM (Soubki et al., 2024)                        | 2024 | written       | –           | explicit belief representation supplied... | ○      |
| self-verification limits (Stechly et al., 2024)         | 2024 | context       | –           | self-critique vs sound external verifie... |        |
| AppWorld (Trivedi et al., 2024)                         | 2024 | context       | –           | none (environment; state-based unit tes... | ○      |
| SOTOPIA-pi (Wang et al., 2024a)                         | 2024 | –             | output      | LLM-rated social goal completion used f... |        |
| Agent Workflow Memory (Wang et al., 2024d)              | 2024 | ext. memory   | –           | induction of reusable workflows from su... |        |
| DistRL (Wang et al., 2024b)                             | 2024 | context       | trajectory  | none (state is screenshot history in co... |        |
| MEMORYLLM (Wang et al., 2024c)                          | 2024 | prob. memory  | state       | new text knowledge injected into latent... | ○      |
| POAD (Wen et al., 2024)                                 | 2024 | context       | interaction | token-level Bellman backup (BAD) assign... |        |
| Belief-R (Wilie et al., 2024)                           | 2024 | context       | –           | new premise that should defeat prior in... | ✓      |
| LongMemEval (Wu et al., 2024)                           | 2024 | ext. memory   | –           | knowledge-update questions test whether... | ✓      |
| AgentGym (Xi et al., 2024)                              | 2024 | context       | trajectory  | AgentEvol: iterative exploration + SFT...  |        |
| TradingAgents (Xiao et al., 2024)                       | 2024 | context       | output      | multi-agent debate (bull vs bear resear... |        |
| tau-bench (Yao et al., 2024)                            | 2024 | –             | –           | simulated-user utterances and API tool...  |        |
| LaBToM (Ying et al., 2024)                              | 2024 | ext. filter   | –           | Bayesian inverse planning over the obse... | ✓      |
| FinCon (Yu et al., 2024)                                | 2024 | ext. memory   | state       | episodic self-critique producing concep... |        |
| FinAgent (Zhang et al., 2024a)                          | 2024 | ext. memory   | interaction | low-level and high-level reflection ove... |        |
| MemSim (Zhang et al., 2024c)                            | 2024 | ext. memory   | –           | none (evaluation simulator)                | ○      |
| ArCHer (Zhou et al., 2024b)                             | 2024 | –             | interaction | hierarchical RL: utterance-level off-po... |        |
| Reflect-RL (Zhou et al., 2024a)                         | 2024 | written       | interaction | frozen reflection model produces natura... |        |
| Beliefs of self and others (Zhu et al., 2024)           | 2024 | context       | –           | none (probing and activation interventi... | ✓      |
| Bayesian geometry of attention (Agarwal et al., 2025)   | 2025 | context       | –           | implicit posterior update via FFN in HM... | ✓      |
| Markovian Thinker (Aghajohari et al., 2025)             | 2025 | written       | output      | none                                       |        |
| MemoryBench (Ai et al., 2025)                           | 2025 | ext. memory   | –           | accumulated simulated user feedback dur... | ○      |
| BASIL (Atwell et al., 2025)                             | 2025 | context       | output      | user pushback vs evidence; Bayesian-con... | ✓      |
| Vending-Bench (Backlund and Petersson, 2025)            | 2025 | ext. memory   | –           | none (evaluation only; agent has scratc... | ○      |
| Align While Search (Bae et al., 2025)                   | 2025 | ext. filter   | –           | posterior update from action-conditione... | ✓      |
| tau2-bench (Barres et al., 2025)                        | 2025 | –             | –           | user actions on shared environment obse... |        |
| LOOP (Chen et al., 2025c)                               | 2025 | context       | trajectory  | environment feedback via API observatio... |        |
| ReSearch (Chen et al., 2025d)                           | 2025 | context       | output      | search results inserted into reasoning...  |        |
| HaluMem (Chen et al., 2025b)                            | 2025 | ext. memory   | –           | memory extraction and memory updating o... | ✓      |
| SPA (Chen et al., 2025e)                                | 2025 | world model   | WM loss     | self-experience SFT supervises current...  | ○      |
| GUI-Shepherd (Chen et al., 2025a)                       | 2025 | –             | interaction | process reward model trained on 52k hum... |        |
| Agent-R1 (Cheng et al., 2025)                           | 2025 | context       | trajectory  | tool/environment feedback per interacti... |        |
| Mem0 (Chhikara et al., 2025)                            | 2025 | ext. memory   | –           | LLM-driven extraction, consolidation an... |        |
| Debate or Vote (Choi et al., 2025)                      | 2025 | context       | output      | peer arguments in multi-agent debate ro... | ○      |
| Language and Experience (Colas et al., 2025)            | 2025 | world model   | –           | Bayesian inference over executable worl... | ○      |
| ARPO (Dong et al., 2025a)                               | 2025 | context       | trajectory  | token entropy spike after tool feedback... |        |
| Active Confusion Expression (Du et al., 2025)           | 2025 | written       | output      | confusion indicators (e.g. 'tricky'/'co... | ○      |
| SoMi-ToM (Fan et al., 2025)                             | 2025 | context       | –           | new first-person multimodal observation... | ○      |
| GiGPO (Feng et al., 2025b)                              | 2025 | –             | trajectory  | episode-level group reward plus step-le... |        |
| ReTool (Feng et al., 2025a)                             | 2025 | context       | trajectory  | outcome reward only; no per-step or sta... |        |
| AgentRefine (Fu et al., 2025)                           | 2025 | context       | trajectory  | environment observation triggers correc... |        |
| Survey of self-evolving agents (Gao et al., 2025)       | 2025 | –             | –           | none                                       |        |
| AccumulatingContext (Geng et al., 2025)                 | 2025 | context       | –           | beliefs drift as context accumulates th... | ✓      |
| DIAMONDS (Ghosh et al., 2025)                           | 2025 | –             | –           | values of tracked quantities change ove... | ✓      |
| GenEnv (Guo et al., 2025)                               | 2025 | world model   | trajectory  | alpha-Curriculum Reward aligns simulato... |        |
| Martingale Score (He et al., 2025)                      | 2025 | context       | –           | new evidence across iterative reasoning... | ✓      |
| Measuring Sycophancy of Language... (Hong et al., 2025) | 2025 | context       | –           | sustained user pressure across turns       | ○      |
| MemoryAgentBench (Hu et al., 2025c)                     | 2025 | –             | –           | incremental multi-turn information inje... | ✓      |
| Memory in the Age of AI Agents (Hu et al., 2025d)       | 2025 | ext. memory   | –           | memory formation, evolution and retriev... |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 19 | Master table, continued (4/10).**

| System                                              | Year | State kept by | Credited    | Revision trigger                           | Metric |
|-----------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| Agent2World (Hu et al., 2025b)                      | 2025 | world model   | trajectory  | testing-team unit tests and simulation-... | ○      |
| Text2World (Hu et al., 2025a)                       | 2025 | world model   | output      | none (one-shot world-model generation,...  | ○      |
| Bayesian Coherence Coefficient (Imran et al., 2025) | 2025 | context       | –           | in-context evidence pieces presented wi... | ✓      |
| Search-R1 (Jin et al., 2025b)                       | 2025 | context       | output      | retrieved documents interleaved with re... |        |
| SearchRLStudy (Jin et al., 2025a)                   | 2025 | context       | trajectory  | outcome reward plus format reward; inte... |        |
| Emergent World Beliefs (Kamel et al., 2025)         | 2025 | world model   | –           | none (probing of pretrained activations)   | ✓      |
| ACON (Kang et al., 2025b)                           | 2025 | written       | –           | compression guidelines refined in natur... | ○      |
| MemoryOS (Kang et al., 2025a)                       | 2025 | ext. memory   | –           | dialogue-chain FIFO from short- to mid-... | ○      |
| CoEx (Kim and won Hwang, 2025)                      | 2025 | world model   | –           | subgoal experience assimilated into neu... |        |
| RefAct (Kim et al., 2025)                           | 2025 | written       | –           | each step rewrites a natural-language s... | ○      |
| LLMs get lost multi-turn (Laban et al., 2025)       | 2025 | context       | –           | new user information across turns (mode... | ○      |
| Hindsight (Latimer et al., 2025)                    | 2025 | ext. memory   | –           | reflect operation updates evolving beli... |        |
| CWM games (Lehrach et al., 2025)                    | 2025 | world model   | –           | unclear (model synthesised from rules a... | ○      |
| DeepAgent (Li et al., 2025e)                        | 2025 | ext. memory   | interaction | autonomous memory folding compresses in... |        |
| Word to World (Li et al., 2025f)                    | 2025 | world model   | WM loss     | next-state prediction under interaction    | ○      |
| Turn-PPO (Li et al., 2025c)                         | 2025 | context       | interaction | turn-level advantage estimation in turn... |        |
| SALT (Li et al., 2025b)                             | 2025 | –             | interaction | none                                       |        |
| SORL (Li et al., 2025a)                             | 2025 | context       | interaction | turn-level importance sampling; clippin... |        |
| MemOS (Li et al., 2025g)                            | 2025 | ext. memory   | –           | memory lifecycle scheduling: MemCubes w... |        |
| AgentFlow (Li et al., 2025h)                        | 2025 | ext. memory   | trajectory  | verifier output appended to evolving me... |        |
| RecToM (Li et al., 2025d)                           | 2025 | –             | –           | questions require tracking evolving int... | ○      |
| ABELL (Lidayan et al., 2025)                        | 2025 | written       | state       | belief grading: ground-truth grader or...  | ○      |
| iStar (Liu et al., 2025)                            | 2025 | context       | interaction | implicit PRM trained by trajectory-leve... |        |
| M3-Agent (Long et al., 2025)                        | 2025 | ext. memory   | trajectory  | streaming visual/auditory input continu... | ○      |
| ARPO (Lu et al., 2025)                              | 2025 | context       | trajectory  | none                                       |        |
| Agent Lightning (Luo et al., 2025b)                 | 2025 | context       | interaction | credit assignment module decomposes tra... |        |
| UltraHorizon (Luo et al., 2025a)                    | 2025 | context       | –           | none                                       | ○      |
| R-WoM (Mei et al., 2025)                            | 2025 | world model   | –           | retrieved tutorial knowledge grounds si... | ○      |
| MATPO (Mo et al., 2025)                             | 2025 | –             | trajectory  | credit assigned across planner and work... |        |
| MindGames (Moore et al., 2025)                      | 2025 | –             | –           | agent must ask about the interlocutor's... | ○      |
| LLaMAR (Nayak et al., 2024)                         | 2025 | context       | –           | none                                       |        |
| WMNav (Nie et al., 2025)                            | 2025 | world model   | –           | discrepancy between world-model plan an... | ○      |
| ReasoningBank (Ouyang et al., 2025)                 | 2025 | ext. memory   | –           | self-judged success/failure of each tas... | ○      |
| ID-RAG (Platnick et al., 2025)                      | 2025 | ext. memory   | –           | unclear (identity KG queried each decis... | ○      |
| Lookbacks belief tracking (Prakash et al., 2025)    | 2025 | context       | –           | visibility lookback updates observing c... | ○      |
| LLM Debate Overconfidence (Prasad and Nguyen, 2025) | 2025 | context       | –           | new opponent arguments each round (shou... | ✓      |
| ToolIRL (Qian et al., 2025)                         | 2025 | context       | output      | reward on tool-call format and correctn... |        |
| Bayesian teaching (Qiu et al., 2026a)               | 2025 | written       | state       | distillation of a normative Bayesian mo... | ✓      |
| Zep / Graphiti (Rasmussen et al., 2025)             | 2025 | ext. memory   | –           | temporally-aware KG edge invalidation w... | ○      |
| Cognitive Memory survey (Shan et al., 2025)         | 2025 | ext. memory   | –           | none (survey; covers updating, conflict... |        |
| TTI (Thinking vs Doing) (Shen et al., 2025)         | 2025 | context       | trajectory  | additional environment observations fro... |        |
| ToMATO (Shinoda et al., 2025)                       | 2025 | context       | –           | information asymmetry between role-play... | ✓      |
| WMACT (Shu et al., 2025)                            | 2025 | context       | output      | none (outcome reward)                      |        |
| R4 (Sohn et al., 2025)                              | 2025 | world model   | –           | continuous 4D database construction fro... |        |
| R1-Searcher (Song et al., 2025)                     | 2025 | context       | trajectory  | none (search results appended to contex... |        |
| ContextFolding (Sun et al., 2025b)                  | 2025 | written       | memory op   | agent branches into a sub-trajectory an... |        |
| ZeroSearch (Sun et al., 2025a)                      | 2025 | context       | trajectory  | none (outcome reward on final answer; s... |        |
| Dynamic Cheatsheet (Suzgun et al., 2025)            | 2025 | ext. memory   | memory op   | self-curation after each query: model w... | ○      |
| TARL (Tan et al., 2025b)                            | 2025 | context       | interaction | LLM-judge turn-level adjudication          |        |
| MemBench (Tan et al., 2025a)                        | 2025 | ext. memory   | –           | none (evaluation only)                     | ○      |
| LLM cooperative game simulation (Tao et al., 2025)  | 2025 | ext. filter   | –           | Bayesian belief updating plus noisy rep... | ○      |
| NextLat belief-state latents (Teoh et al., 2025)    | 2025 | world model   | WM loss     | auxiliary next-latent prediction loss;...  | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 20 | Master table, continued (5/10).**

| System                                                     | Year | State kept by | Credited    | Revision trigger                           | Metric |
|------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| RAGEN / StarPO (Wang et al., 2025f)                        | 2025 | context       | trajectory  | environment reward; reasoning-aware rew... |        |
| VAGEN (Wang et al., 2025d)                                 | 2025 | written       | state       | world-modeling reward on turn-level sta... | ○      |
| IGPO (Wang et al., 2025a)                                  | 2025 | context       | interaction | turn-level information gain: marginal i... | ○      |
| ENACT (Wang et al., 2025e)                                 | 2025 | world model   | –           | none (evaluation only)                     | ○      |
| EMPG (Wang et al., 2025c)                                  | 2025 | context       | interaction | step-wise policy entropy modulates grad... | ○      |
| Memento 2 stateful reflective mem... (Wang, 2025)          | 2025 | ext. memory   | memory op   | memory write = policy evaluation, read...  |        |
| SPA-RL (Wang et al., 2025b)                                | 2025 | –             | interaction | none                                       |        |
| StepSearch (Wang et al., 2025g)                            | 2025 | context       | interaction | per-step information gain and redundanc... |        |
| Fine-Grained Reward Credit (Wei et al., 2025a)             | 2025 | context       | interaction | per-turn environment rewards in turn-le... |        |
| WebAgent-R1 (Wei et al., 2025d)                            | 2025 | context       | trajectory  | none (binary task-success reward only)     |        |
| GTR-Turbo (Wei et al., 2025c)                              | 2025 | –             | trajectory  | none                                       |        |
| Evo-Memory (Wei et al., 2025b)                             | 2025 | ext. memory   | –           | memory refined after each task in a seq... | ○      |
| ReSum (Wu et al., 2025c)                                   | 2025 | written       | trajectory  | periodic tool-based summarization when...  |        |
| RLVR-World (Wu et al., 2025a)                              | 2025 | world model   | WM loss     | RL reward on decoded transition-predict... | ○      |
| CollabLLM (Wu et al., 2025b)                               | 2025 | context       | interaction | none for belief; Multiturn-aware Reward... |        |
| DEL-ToM (Wu et al., 2025d)                                 | 2025 | written       | state       | each DEL belief-update step scored by a... | ✓      |
| AgentPRM (Xi et al., 2025)                                 | 2025 | context       | interaction | TD/GAE-estimated promise and progress l... |        |
| Experience-following memory study (Xiong et al., 2025b)    | 2025 | ext. memory   | memory op   | addition gated by trajectory evaluators... | ○      |
| FLAG-TRADER (Xiong et al., 2025a)                          | 2025 | context       | interaction | policy-gradient RL on trading reward ov... |        |
| A-Mem (Xu et al., 2025e)                                   | 2025 | ext. memory   | output      | memory evolution: new notes trigger upd... |        |
| STARE-VLA (Xu et al., 2025a)                               | 2025 | –             | interaction | none                                       |        |
| PPR (Xu et al., 2025c)                                     | 2025 | context       | trajectory  | none                                       |        |
| EnigmaToM (Xu et al., 2025b)                               | 2025 | written       | –           | iterative masking over entity-state sce... | ○      |
| Memory-R1 (Yan et al., 2025)                               | 2025 | ext. memory   | memory op   | learned ADD/UPDATE/DELETE/NOOP operatio... |        |
| EvolvTrip (Yan et al., 2025b)                              | 2025 | written       | –           | narrative events append to a perspectiv... | ○      |
| ToM-agent counterfactual reflecti... (Yang et al., 2025a)  | 2025 | written       | –           | counterfactual gap between predicted an... | ○      |
| ECON (Yi et al., 2025)                                     | 2025 | ext. filter   | trajectory  | TD-loss update of a learned belief netw... |        |
| MemAgent (Yu et al., 2025b)                                | 2025 | written       | state       | chunk-wise overwrite of fixed-size memo... |        |
| Dyna-Think (Yu et al., 2025c)                              | 2025 | world model   | WM loss     | DDT stage 1 trains state prediction / s... | ✓      |
| Demystifying agentic RL (Yu et al., 2025d)                 | 2025 | context       | trajectory  | outcome reward on end-to-end tool-use t... |        |
| PersuasiveToM (Yu et al., 2025a)                           | 2025 | –             | –           | questions probe shifts of desires/belie... | ○      |
| EvolveCast (Yuan et al., 2025b)                            | 2025 | context       | –           | post-cutoff evidence presented; forecas... | ✓      |
| MARSHAL (Yuan et al., 2025a)                               | 2025 | –             | interaction | turn-level advantage estimator aligned...  |        |
| Agentic RL survey (Zhang et al., 2025d)                    | 2025 | –             | –           | n/a                                        |        |
| CriticSearch (Zhang et al., 2025i)                         | 2025 | –             | interaction | none                                       |        |
| LLMs as discounted Bayesian filte... (Zhang et al., 2025e) | 2025 | context       | –           | sequential stochastic observations unde... | ✓      |
| ACE (Zhang et al., 2025g)                                  | 2025 | written       | –           | reflector extracts lessons from executi... | ○      |
| Memory in LLMs: mechanisms/eval (Zhang et al., 2025c)      | 2025 | ext. memory   | –           | write->read->inhibit/update chain; mode... | ○      |
| ReasonRAG (Zhang et al., 2025j)                            | 2025 | context       | interaction | process-level rewards on query generati... |        |
| Ark / MedArk (Zhang et al., 2025a)                         | 2025 | context       | trajectory  | ask-vs-tell decision via self-reasoning... | ○      |
| MindPower (Zhang et al., 2025h)                            | 2025 | written       | output      | Mind-Reward rewards consistency between... | ○      |
| Beyond Markovian (Zhang et al., 2025i)                     | 2025 | context       | trajectory  | observed outcomes update a posterior ov... |        |
| PRIME (Zhang et al., 2025k)                                | 2025 | ext. memory   | –           | new user engagements update episodic me... |        |
| Sycophancy under Pressure (Zhang et al., 2025f)            | 2025 | context       | output      | adversarial user pressure in dialogue;...  | ○      |
| Overcoming Multi-step Complexity... (Zhang et al., 2025b)  | 2025 | ext. filter   | output      | stepwise Bayesian updates with small-LM... | ○      |
| PrefEval (Zhao et al., 2025)                               | 2025 | context       | –           | user preference stated explicitly or im... | ○      |
| DeepResearcher (Zheng et al., 2025c)                       | 2025 | context       | trajectory  | outcome reward on final answer from end... | ○      |
| LifelongAgentBench (Zheng et al., 2025b)                   | 2025 | ext. memory   | –           | cross-task experience replay; group sel... | ○      |
| MEM1 (Zhou et al., 2025d)                                  | 2025 | written       | state       | internal state rewritten each turn cons... |        |
| Memento (Zhou et al., 2025a)                               | 2025 | ext. memory   | interaction | environment feedback rewrites episodic...  |        |
| SWEET-RL (Zhou et al., 2025c)                              | 2025 | –             | interaction | step-level critic reward using training... |        |
| Social World Models (S3AP) (Zhou et al., 2025b)            | 2025 | written       | –           | structured state/action/mental-state re... | ○      |

*Metric:* ✓ belief-level metric reported, ○ proxy only.

**Table 21 | Master table, continued (6/10).**

| System                                                            | Year | State kept by | Credited    | Revision trigger                            | Metric |
|-------------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| T3 (Zou et al., 2025)                                             | 2025 | prob. memory  | trajectory  | truth-anchored potential -log b(s*) sta...  | ✓      |
| MEDLEY-BENCH (Abtahi et al., 2026)                                | 2026 | context       | –           | social disagreement / analyst summaries...  |        |
| Agent-ToM: learning to monitor ag... (Ahmed and Nafisi, 2026)     | 2026 | –             | trajectory  | deviation from task-consistent expected...  |        |
| Can LLM Agents Price Competitivel... (Ahmed et al., 2026)         | 2026 | context       | –           | market feedback and demand shocks revis...  | ○      |
| Bayesian Chronicle Agents (BCA) (Akbar et al., 2026)              | 2026 | ext. filter   | –           | Bayesian update (Beta pseudo-counts, pr...  | ✓      |
| Why world models: LLM state-track... (Alaswad et al., 2026)       | 2026 | context       | –           | none                                        |        |
| RLAW: POMDP routing + self-correc... (Ali and Tseng, 2026)        | 2026 | ext. memory   | interaction | critique score (logic/safety check)         |        |
| Bayesian orchestration (Amin, 2026)                               | 2026 | ext. filter   | state       | Bayes-rule posterior update with LLM-el...  | ○      |
| Clue possibility matrix (Ansell and Toney-Wails, 2026)            | 2026 | ext. filter   | –           | symbolic constraint update (YES/NO/MAYB...  | ✓      |
| CA-BED (Arnould et al., 2026)                                     | 2026 | prob. memory  | –           | LLM-estimated likelihoods of user answe...  | ○      |
| RoVLaP (Azran et al., 2026)                                       | 2026 | ext. filter   | –           | VLM next-token predicate probabilities...   |        |
| BAG (Baan et al., 2026)                                           | 2026 | context       | –           | user clarification in the next turn res...  | ○      |
| RB-VLA: recursive belief VLA (Bagaria et al., 2026)               | 2026 | world model   | state       | self-supervised world-model objective (...) | ○      |
| PatchWorld (Bai et al., 2026)                                     | 2026 | world model   | WM loss     | counterexample-guided repair of executa...  | ○      |
| LLM + causal model POMDP planning (Basu et al., 2026)             | 2026 | ext. filter   | –           | POMDP observation update; hybrid confid...  | ○      |
| OmniToM explicit belief modeling (Bawatneh et al., 2026)          | 2026 | –             | –           | n/a                                         | ✓      |
| MIST (Bensal et al., 2026)                                        | 2026 | ext. memory   | –           | user misconceptions written into memory...  | ✓      |
| ECCBench (Berman and Deng, 2026)                                  | 2026 | –             | –           | none                                        | ✓      |
| CAGE-2 compound agent design (Bogdanov et al., 2026)              | 2026 | ext. memory   | –           | deterministic state-tracking layer with...  |        |
| Reasoning Theater (Boppa et al., 2026)                            | 2026 | context       | output      | internal activation belief shift during...  | ○      |
| InKH financial harness (Borjigin et al., 2026)                    | 2026 | ext. memory   | –           | passive knowledge injection from user/m...  |        |
| CPS statistical control survey (Cai et al., 2026)                 | 2026 | –             | –           | none                                        |        |
| TRACE (Chang and Chang, 2026)                                     | 2026 | written       | –           | append-only revision when world-model c...  |        |
| Trivium (Chang, 2026)                                             | 2026 | world model   | WM loss     | budgeted probes and continued posterior...  | ✓      |
| STALE (Chao et al., 2026)                                         | 2026 | ext. memory   | –           | implicit conflict                           | ✓      |
| Belief-State Engine (Chattopadhyay and Halder, 2026)              | 2026 | ext. filter   | –           | Bayesian posterior                          | ✓      |
| The Horizon Gap (survey) (Chen et al., 2026b)                     | 2026 | –             | –           | n/a                                         |        |
| BCO: belief-calibrated scaffold o... (Chen et al., 2026c)         | 2026 | written       | state       | evaluation of new candidate scaffolds (...) | ✓      |
| MM-Lifelong / ReMA (Chen et al., 2026a)                           | 2026 | ext. memory   | –           | dynamic memory management iteratively u...  | ○      |
| Theory of Agent (ToA) (Cheng et al., 2026)                        | 2026 | –             | –           | none                                        |        |
| Truth stance layer (expressed dou... (Cochinescu, 2026)           | 2026 | prob. memory  | –           | rule-based revision operator (user corr...  | ✓      |
| BOND (distilling Bayesian beliefs) (Cui and Lin, 2026)            | 2026 | written       | state       | teacher posterior                           | ✓      |
| Closing the Feedback Loop (Cui et al., 2026)                      | 2026 | ext. memory   | –           | realised returns interpreted by an LLM...   |        |
| STOCKTAKE (Deb and Krishnan, 2026)                                | 2026 | –             | –           | n/a                                         | ✓      |
| MDLM text world models (Deshpande, 2026)                          | 2026 | world model   | WM loss     | world model conditions on state anchors...  | ○      |
| DeltaLogic minimal-edit revision (Dhanda, 2026)                   | 2026 | context       | output      | minimal premise edit                        | ✓      |
| HIPIF (Diao et al., 2026)                                         | 2026 | written       | trajectory  | subgoal completion (history folding) +...   |        |
| SGCD sibling-guided credit (Ding et al., 2026b)                   | 2026 | –             | output      | none                                        |        |
| APEx (Ding et al., 2026a)                                         | 2026 | ext. memory   | memory op   | reward-guided distillation of trajector...  |        |
| Belief at Risk (Dixon, 2026b)                                     | 2026 | ext. filter   | –           | Bayesian posterior over regimes             | ✓      |
| POMDP validation framework (Dixon, 2026c)                         | 2026 | ext. filter   | –           | Bayesian                                    | ✓      |
| Governance-aware POMDP for AI del... (Dixon, 2026a)               | 2026 | ext. filter   | –           | Bayesian update on evidence quality         |        |
| Mind-Studio (Dong et al., 2026)                                   | 2026 | world model   | –           | entropy-selected state-action-next-stat...  | ✓      |
| LifeSim (Duan et al., 2026)                                       | 2026 | context       | –           | user simulator driven by BDI model; ass...  | ○      |
| KbSD (Feng et al., 2026)                                          | 2026 | context       | output      | none                                        | ○      |
| MedExAgent (Gao et al., 2026a)                                    | 2026 | context       | output      | none (composite outcome reward: accurac...  |        |
| Auditing belief-conditioned Werew... (Gao et al., 2026b)          | 2026 | ext. memory   | state       | logged per-turn belief updates              | ✓      |
| CubeBench (ang Gao et al., 2025)                                  | 2026 | –             | –           | n/a (benchmark)                             | ○      |
| Cyclic subtask graphs (Gharzeddine and Saab, 2026)                | 2026 | written       | –           | state-analysis-and-routing agent applic...  |        |
| FinBench calibration benchmark (Ghosh and Devarakonda, 2026)      | 2026 | –             | output      | none                                        |        |
| CHAL: council of hierarchical age... (Giovannelli and Kent, 2026) | 2026 | written       | state       | gradient-informed revision on thesis st...  |        |
| Does Theory of Mind Improvement R... (Gong et al., 2026)          | 2026 | context       | –           | live user responses in interactive firs...  | ○      |

*Metric:* ✓ belief-level metric reported, ○ proxy only.

**Table 22 | Master table, continued (7/10).**

| System                                                         | Year | State kept by | Credited    | Revision trigger                             | Metric |
|----------------------------------------------------------------|------|---------------|-------------|----------------------------------------------|--------|
| ENIGMA (Gritsaenko, 2026)                                      | 2026 | prob. memory  | –           | deterministic rule-based belief crystal. . . | ✓      |
| HiDiffTIR (Guo et al., 2026c)                                  | 2026 | context       | interaction | group tool-call success rate                 |        |
| PivoARL (Guo et al., 2026b)                                    | 2026 | context       | interaction | structured reflection identifies pivota. . . |        |
| CAVE (Guo et al., 2026a)                                       | 2026 | context       | interaction | process reward decomposed into belief u. . . | ○      |
| EnterpriseArena (Han et al., 2026)                             | 2026 | context       | –           | none                                         |        |
| EVAF (Han, 2026)                                               | 2026 | ext. memory   | memory op   | surprise- and valence-gated consolidati. . . | ○      |
| Structured memory over-personaliz. . . (Hannoon et al., 2026)  | 2026 | ext. memory   | –           | none (inference-time restructuring of m. . . | ○      |
| MemOps (Hao et al., 2026)                                      | 2026 | ext. memory   | –           | lifecycle ops (remember/forget/update/r. . . | ✓      |
| YC-Bench (He et al., 2026b)                                    | 2026 | written       | –           | scratchpad persistence across context t. . . |        |
| TRACE turn credit for jailbreaking (He et al., 2026d)          | 2026 | context       | interaction | leave-one-turn-out semantic masking          |        |
| HGPO (He et al., 2026c)                                        | 2026 | –             | interaction | none                                         |        |
| CoLearn (He et al., 2026a)                                     | 2026 | prob. memory  | –           | Bayesian update (soft-evidence BKT; LLM. . . | ✓      |
| CAPTURE: preference drift vs memo. . . (Hossain et al., 2026)  | 2026 | world model   | state       | uncertainty-triggered clarification + c. . . | ✓      |
| REDEREF: training-free probabilis. . . (Hosseini et al., 2026) | 2026 | prob. memory  | interaction | judge feedback updates Thompson-samplin. . . |        |
| WMF-AM working memory probe (Hou et al., 2026)                 | 2026 | context       | –           | K sequential operations within one quer. . . | ✓      |
| MATTRL test-time multi-agent RL (Hu et al., 2026b)             | 2026 | ext. memory   | interaction | turn-level credit for experience pool c. . . |        |
| BRAID (Hu et al., 2026c)                                       | 2026 | –             | interaction | none                                         |        |
| Infinite-Parameter LLM (Hu et al., 2026a)                      | 2026 | ext. filter   | state       | known answer (gold passage relevance la. . . | ✓      |
| SDPO (Hübotter et al., 2026)                                   | 2026 | context       | output      | textual environment feedback (errors, j. . . |        |
| ToMAgent lookahead ToM (Hwang et al., 2025)                    | 2026 | written       | state       | dialogue lookahead rollouts score (ment. . . | ○      |
| MARBO (Hwang et al., 2026)                                     | 2026 | written       | state       | cross-entropy against ground-truth role. . . | ○      |
| Persuasion Propagation (Jeong et al., 2026)                    | 2026 | context       | –           | none                                         | ○      |
| Planted latent variable (Jerpelea, 2026)                       | 2026 | context       | –           | Bayesian posterior over the planted lat. . . | ✓      |
| EOG belief propagation (Jha et al., 2026)                      | 2026 | ext. filter   | –           | new evidence labelled by the LLM (cause. . . | ○      |
| Hindsight Memory-PRM (Jia et al., 2026)                        | 2026 | ext. memory   | memory op   | hindsight audit trail (retrieval hits,. . .  | ○      |
| SCRIBE (Jiang and Ferraro, 2026)                               | 2026 | context       | interaction | skill-prototype rubric reward on subgoa. . . |        |
| HAGE (Jiang et al., 2026b)                                     | 2026 | ext. memory   | memory op   | RL on downstream task jointly updates e. . . |        |
| PABU (Jiang et al., 2026c)                                     | 2026 | written       | state       | SFT on LLM-synthesised progress and ret. . . |        |
| Jev-Mem (Jiang et al., 2026a)                                  | 2026 | ext. memory   | –           | System-One controller types, links and. . .  |        |
| Regent Chess replication (Joshi, 2026)                         | 2026 | written       | –           | none (elicited belief scored only)           | ✓      |
| SuS strategy-aware surprise (Kashirskiy and Makarov, 2026)     | 2026 | context       | output      | surprise residual (pre-post prediction. . .  |        |
| Context Gathering Decision Process (Kausik et al., 2026)       | 2026 | ext. filter   | –           | n/a                                          |        |
| InfoPO (Kong et al., 2026)                                     | 2026 | context       | interaction | information-gain reward: credit to turn. . . | ○      |
| Belief-Based World Model (Kumar et al., 2026)                  | 2026 | world model   | –           | world-model belief                           |        |
| Hindsight (Latimer et al., 2026)                               | 2026 | ext. memory   | –           | retain/reflect operations; opinions car. . . | ○      |
| MINTEval (Lee et al., 2026a)                                   | 2026 | –             | –           | none (evaluation of memory under update. . . | ✓      |
| Meta-Harness (Lee et al., 2026b)                               | 2026 | ext. memory   | memory op   | outer-loop search over harness code usi. . . |        |
| Belief geometries via SAEs (Levinson, 2026)                    | 2026 | context       | –           | none                                         | ○      |
| Text World Models (review) (Li et al., 2026d)                  | 2026 | world model   | –           | n/a                                          |        |
| ECPO (Li et al., 2026e)                                        | 2026 | –             | trajectory  | n/a                                          |        |
| SERL selective hindsight distilla. . . (Li et al., 2026c)      | 2026 | context       | interaction | per-step environment feedback (error me. . . |        |
| VICT (Li et al., 2026b)                                        | 2026 | –             | interaction | verifier atom proof edges                    |        |
| InfoDelphi (Li et al., 2026f)                                  | 2026 | context       | –           | private evidence surfaced through peer. . .  | ✓      |
| SE-Search (Li et al., 2026a)                                   | 2026 | ext. memory   | memory op   | memory purification at each Think-Searc. . . |        |
| RTPO (Li et al., 2026g)                                        | 2026 | –             | interaction | reverse-turn tree updates align each tu. . . |        |
| ELPO (Liang et al., 2026)                                      | 2026 | –             | interaction | first irrecoverable step located by bin. . . |        |
| Belief Memory (BeliefMem) (Liao et al., 2026a)                 | 2026 | ext. memory   | –           | Noisy-OR update                              |        |
| TCPO turn-level credit (Liao et al., 2026b)                    | 2026 | context       | interaction | verifier score-to-credit conversion; hi. . . |        |
| UQ for LLM Agents taxonomy (Lin et al., 2026c)                 | 2026 | –             | –           | n/a                                          | ✓      |
| SkillC (Lin et al., 2026a)                                     | 2026 | context       | trajectory  | skill-vs-no-skill reward gap                 |        |
| ReAdapt (Lin et al., 2026b)                                    | 2026 | written       | –           | LLM Adapt step after each tool observat. . . |        |
| MADE: belief-driven dual-agent de. . . (Liu et al., 2026e)     | 2026 | ext. memory   | interaction | execution failure -> root-cause invalid. . . |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 23 | Master table, continued (8/10).**

| System                                                        | Year | State kept by | Credited    | Revision trigger                            | Metric |
|---------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| PiCA pivot-based credit (Liu et al., 2026b)                   | 2026 | context       | trajectory  | pivot-step potential-based shaping          |        |
| SHADOW (Liu et al., 2026g)                                    | 2026 | context       | interaction | dynamics-aware state grouping and local...  |        |
| INTENT (Liu et al., 2026c)                                    | 2026 | world model   | –           | inference-time: intention-aware hierarc...  | ○      |
| SimpleMem (Liu et al., 2026d)                                 | 2026 | ext. memory   | –           | online semantic synthesis merges relate...  | ○      |
| WorldMemArena (Liu et al., 2026a)                             | 2026 | ext. memory   | –           | gold memory updates and distractors ann...  | ✓      |
| PRISM (Liu et al., 2026h)                                     | 2026 | written       | –           | new observations and VLM semantic reaso...  |        |
| MMPO (Liu et al., 2026i)                                      | 2026 | written       | state       | belief-entropy proxy (own answer entrop...  |        |
| REE diagnostic for ICL in games (Liu et al., 2026f)           | 2026 | context       | –           | none (interaction history in context)       |        |
| AutoHarness (Lou et al., 2026)                                | 2026 | ext. filter   | –           | environment feedback (illegal-move erro...  | ○      |
| HISR (Lu et al., 2026c)                                       | 2026 | context       | interaction | segment-level process reward modulated...   |        |
| UI-Copilot (Lu et al., 2026b)                                 | 2026 | ext. memory   | interaction | memory decoupling separates persistent...   |        |
| PaW (Lu et al., 2026a)                                        | 2026 | world model   | WM loss     | clipped-MAE auxiliary loss on next-obse...  | ○      |
| From Storage to Experience (memor... (Luo et al., 2026c)      | 2026 | ext. memory   | –           | reflection                                  |        |
| Graph of States (Luo et al., 2026d)                           | 2026 | ext. memory   | –           | symbolic check (causal graph constraint...  |        |
| ReflectiChain world-model supply... (Luo, 2026)               | 2026 | world model   | trajectory  | latent trajectory rehearsals; reflection... |        |
| ReflectiChain (Luo et al., 2026a)                             | 2026 | world model   | –           | retrospective reflection over the execu...  | ✓      |
| PUMA user-state modeling (Luo et al., 2026b)                  | 2026 | prob. memory  | WM loss     | new user utterance updates variational...   | ✓      |
| ProPlay (Ma et al., 2026b)                                    | 2026 | world model   | trajectory  | post-episode graph refinement from envi...  |        |
| Fine-Mem (Ma et al., 2026a)                                   | 2026 | ext. memory   | memory op   | fine-grained feedback alignment on memo...  |        |
| LongHorizon-Harness (Ma et al., 2026c)                        | 2026 | ext. memory   | –           | none                                        |        |
| CoMAM (Mao et al., 2026b)                                     | 2026 | ext. memory   | memory op   | none                                        |        |
| TreeMem (Mao et al., 2026a)                                   | 2026 | ext. memory   | memory op   | none                                        |        |
| Few-shot in-context world represe... (Matsutani et al., 2026) | 2026 | context       | –           | none (in-context inference)                 | ✓      |
| Physics of long-horizon planning (Men et al., 2026)           | 2026 | world model   | WM loss     | simulator ground-truth state transition...  | ○      |
| EquiMem (Meng et al., 2026)                                   | 2026 | ext. memory   | –           | each memory update scored against equil...  | ○      |
| PCCC / CVT-RL (Meng, 2026)                                    | 2026 | written       | state       | verifier-estimated prefix-observable la...  | ✓      |
| Episodic-Semantic Memory (Milosevic, 2026)                    | 2026 | ext. memory   | memory op   | consolidation of episodic window into s...  | ○      |
| MT-GRPO Iterative Reward Calibrat... (Modercua et al., 2026)  | 2026 | context       | interaction | none (per-turn reward calibrated from r...  |        |
| Bayesian Linguistic Forecaster (Murphy, 2026)                 | 2026 | written       | –           | sequential update in tool loop              | ✓      |
| BeliefShift (Myakala et al., 2026)                            | 2026 | –             | –           | evidence-driven revision                    | ✓      |
| When State Constraints Go Uncheck... (Nakayashiki, 2026)      | 2026 | ext. memory   | –           | budgeted verification                       |        |
| DToM-Track: dynamic ToM as tempor... (Nguyen et al., 2026)    | 2026 | context       | –           | none                                        | ✓      |
| Light-Omni (Nie et al., 2026)                                 | 2026 | ext. memory   | –           | hierarchical merging consolidates episo...  |        |
| EvoHarness-RL learned harness sta... (Ning et al., 2026)      | 2026 | ext. memory   | interaction | learned read/update/consolidate policy...   |        |
| SSMG-Nav (Niu et al., 2026)                                   | 2026 | ext. memory   | –           | new observations consolidated into a pe...  | ○      |
| SkillIOS (Ouyang et al., 2026)                                | 2026 | ext. memory   | memory op   | trainable curator updates SkillRepo fro...  |        |
| MD5 state tracking (Pai and Xian, 2026)                       | 2026 | context       | –           | majority vote over arithmetic               |        |
| Business Arena marketplace benchm... (Pan et al., 2026a)      | 2026 | –             | interaction | none                                        |        |
| ContextPilot (Pan et al., 2026b)                              | 2026 | context       | memory op   | context and entropy variation identify...   |        |
| Bayes-consistent orchestration (Papamarkou et al., 2026)      | 2026 | ext. filter   | –           | Bayesian posterior update over task-rel...  |        |
| Kumiho (Park, 2026)                                           | 2026 | ext. memory   | –           | AGM-style revision on graph memory (imm...  | ○      |
| HiPER: hierarchical RL with expli... (Peng et al., 2026a)     | 2026 | context       | trajectory  | none (reward-based)                         |        |
| Explore More Drift Less (Pu et al., 2026)                     | 2026 | –             | output      | n/a                                         |        |
| STAPO trajectory-neglect steps (Qi et al., 2026)              | 2026 | context       | trajectory  | normalized-entropy outlier steps            |        |
| Agents fail to use world models (Qian et al., 2026)           | 2026 | world model   | –           | agent decides when to invoke external s...  | ○      |
| MoM / P-Mem (Qin and Lu, 2026)                                | 2026 | written       | –           | LLM typed write-time operations (accept...  | ✓      |
| SWIRL (Qiu et al., 2026b)                                     | 2026 | world model   | WM loss     | alternating GRPO updates of forward wor...  | ○      |
| MatchTIR (Qu et al., 2026)                                    | 2026 | context       | interaction | bipartite matching of predicted vs grou...  |        |
| Debugging code world models (Rahmani, 2026)                   | 2026 | world model   | WM loss     | predicted runtime state after each exec...  | ✓      |
| Oblivion (Rana et al., 2026)                                  | 2026 | ext. memory   | memory op   | read gated by agent uncertainty and buf...  |        |
| LLM explainer over Active Inferen... (Raval et al., 2026)     | 2026 | ext. filter   | –           | variational (Active Inference) postero...   | ✓      |
| IAPO: influence-aware credit for... (Ren et al., 2026a)       | 2026 | context       | interaction | none (influence-dependency graph from r...  |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 24 | Master table, continued (9/10).**

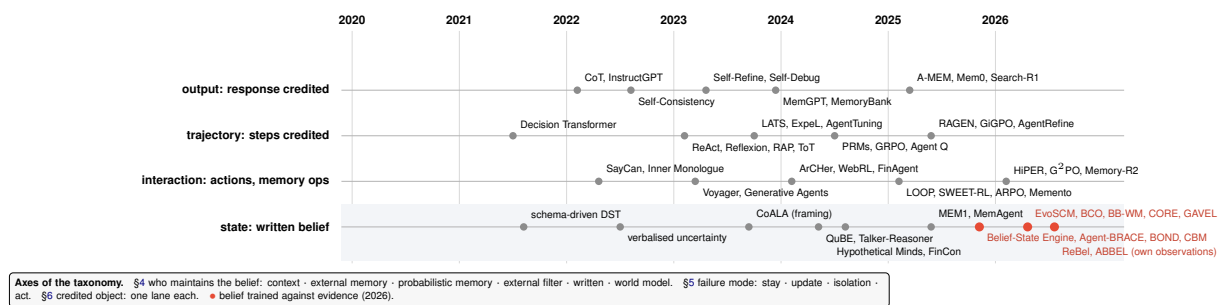
| System                                                      | Year | State kept by | Credited    | Revision trigger                           | Metric |
|-------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| CSSP (Ren et al., 2026b)                                    | 2026 | ext. filter   | state       | new telemetry events update a learned s... | ○      |
| AI Scientists Reasoning (Ríos-García et al., 2026)          | 2026 | context       | –           | refutation-driven revision (rarely obse... | ✓      |
| BayesBench (Samanta et al., 2026)                           | 2026 | context       | –           | per-turn evidence arrival; compared aga... | ✓      |
| PCE (Seo et al., 2026)                                      | 2026 | written       | state       | scenario likelihood scoring                |        |
| FinalityBench (Sharma, 2026)                                | 2026 | context       | –           | none (benchmark; agent probes/waits)       |        |
| SPADER (Shi et al., 2026)                                   | 2026 | context       | interaction | none                                       |        |
| Agent-BRACE (Singh et al., 2026b)                           | 2026 | written       | state       | verbalized certainty                       | ✓      |
| Nous: when belief-based memory he... (Singh, 2026)          | 2026 | prob. memory  | state       | surprise (information-theoretic) + reli... | ✓      |
| MMA-RAG*T (Singh et al., 2026a)                             | 2026 | written       | –           | new observations at each pipeline check... | ○      |
| Pinductor: LLM-prior POMDP world... (Six et al., 2026)      | 2026 | world model   | state       | belief-based likelihood score of candid... |        |
| Parallel WebBench (Sogani et al., 2026)                     | 2026 | context       | trajectory  | GRPO on task outcome; trace diagnostics... | ○      |
| D-MEM (Song and Xin, 2026)                                  | 2026 | ext. memory   | –           | reward prediction error gating             |        |
| Hybrid-WM (Song and Cai, 2026a)                             | 2026 | world model   | state       | consistency gate: disagreement between...  | ✓      |
| WM-SAR (Song and Cai, 2026b)                                | 2026 | world model   | trajectory  | spectral error amplification               |        |
| Semantic Bayesian World Models (Soru, 2026)                 | 2026 | ext. filter   | state       | Bayesian conditioning on observations;...  |        |
| CMI (Srivastava, 2026b)                                     | 2026 | ext. memory   | state       | counterfactual answer-score intervention   |        |
| memory-bench construction (Srivastava, 2026a)               | 2026 | ext. memory   | –           | hidden ledger of facts defines expected... | ✓      |
| External affective state dynamics (Subaharan, 2026)         | 2026 | ext. memory   | –           | exponential smoothing or momentum updat... |        |
| When Memory Updates but Behavior... (Sun and He, 2026)      | 2026 | ext. memory   | –           | state dependency repair                    |        |
| VerMem (Sun et al., 2026b)                                  | 2026 | ext. memory   | memory op   | local verifier scores each memory trans... | ○      |
| AEWM / EditAct (Sun et al., 2026a)                          | 2026 | world model   | interaction | teacher or judge (annotation-agent labe... |        |
| HCAPO hindsight critic (Tan et al., 2026)                   | 2026 | context       | trajectory  | post-hoc LLM critic refinement             |        |
| MSCE (Tang et al., 2026a)                                   | 2026 | ext. memory   | trajectory  | reflection-weighted value backfilling p... | ○      |
| ReBel (Tang et al., 2026b)                                  | 2026 | written       | state       | belief-consistency reward: each predica... | ○      |
| Proactive Service Agents survey (Tang et al., 2026c)        | 2026 | –             | –           | Bayesian belief update (formal framewor... |        |
| STAGE (Tian et al., 2026)                                   | 2026 | written       | state       | new scene evidence triggers recursive u... | ✓      |
| Memora (Uddin et al., 2026)                                 | 2026 | ext. memory   | –           | user-stated add/update/delete of prefer... | ✓      |
| Rank-Bounded Memory (Verbovoy, 2026)                        | 2026 | ext. memory   | –           | storage schema invariants (ownership pa... | ✓      |
| TOKI (Wang, 2026b)                                          | 2026 | ext. memory   | –           | bitemporal operators (4 heuristics)        |        |
| CAPO: critic-guided action-aligne... (Wang et al., 2026a)   | 2026 | context       | interaction | none (critic value at action boundary)     |        |
| SLEA-RL: step-level experience au... (Wang and Jiang, 2026) | 2026 | ext. memory   | trajectory  | observation change triggers step-level...  |        |
| G2PO (Wang et al., 2026j)                                   | 2026 | context       | trajectory  | globally standardized TD error over sta... |        |
| Self-Guide internal reward co-evo... (Wang et al., 2026g)   | 2026 | written       | trajectory  | self-generated guidance as internal rew... |        |
| TAMTRL (Wang et al., 2026d)                                 | 2026 | written       | trajectory  | teacher likelihood of memory update        |        |
| T2PO (Wang et al., 2026b)                                   | 2026 | context       | interaction | token-level entropy/confidence plateau     |        |
| CrEST (Wang et al., 2026k)                                  | 2026 | context       | interaction | verified turn reward with self-teacher...  |        |
| MINDGAMES live arena (Wang et al., 2026c)                   | 2026 | –             | –           | n/a                                        |        |
| APPO (Wang et al., 2026h)                                   | 2026 | –             | output      | none                                       |        |
| LENS (Wang et al., 2026f)                                   | 2026 | prob. memory  | –           | LLM relevance-oracle verdicts update a...  | ○      |
| Mind2Dialogue (Wang et al., 2026l)                          | 2026 | written       | output      | simulator updates a shared evolving use... | ○      |
| Dark Room (Wang, 2026a)                                     | 2026 | written       | state       | rule-verified next-observation predicti... | ○      |
| Conv-FinRe (Wang et al., 2026i)                             | 2026 | –             | –           | n/a (benchmark; investor risk preferenc... | ○      |
| GAVEL (Wang et al., 2026e)                                  | 2026 | ext. filter   | state       | known answer (RSN prior over object roo... | ✓      |
| Agentic Reasoning survey (Wei et al., 2026)                 | 2026 | –             | –           | n/a                                        |        |
| AC1 self-model loop replication (Wilson, 2026)              | 2026 | ext. filter   | –           | Bayesian accumulator over own interacti... | ✓      |
| Bayesian-Agent (Wu et al., 2026b)                           | 2026 | ext. filter   | –           | verified trajectories                      |        |
| BayesEvolve: explicit belief for... (Wu et al., 2026d)      | 2026 | ext. filter   | state       | posterior refit after each evaluated ca... | ✓      |
| HumanLM state alignment (Wu et al., 2026a)                  | 2026 | written       | state       | RL aligns generated natural-language la... | ○      |
| RL recipe long-horizon tool agents (Wu et al., 2026c)       | 2026 | context       | trajectory  | staged vs dense reward shaping on Trave... |        |
| REFLEX (Jev) (Wu and Lim, 2026)                             | 2026 | context       | –           | confidence threshold escalates bounded...  |        |
| SkillRL (Xia et al., 2026)                                  | 2026 | ext. memory   | trajectory  | experience distilled into hierarchical...  |        |
| WebWorld (Xiao et al., 2026b)                               | 2026 | world model   | WM loss     | trained on 1M+ open-web interactions; u... | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 25 | Master table, continued (10/10).**

| System                                                     | Year | State kept by | Credited    | Revision trigger                            | Metric |
|------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| AeroBelief (Xiao et al., 2026a)                            | 2026 | prob. memory  | –           | rule-based accumulation of VLM match co...  |        |
| TIPS (Xie et al., 2026)                                    | 2026 | context       | interaction | teacher answer-likelihood gain              |        |
| LLM Agents for Forecasting (surve... (Xu et al., 2026d)    | 2026 | –             | –           | n/a                                         |        |
| Contextual Belief Management (Bel... (Xu et al., 2026a)    | 2026 | written       | state       | symbolic verification                       | ✓      |
| NeSyFS: neuro-symbolic fast-slow... (Xu and Fekri, 2026)   | 2026 | ext. memory   | interaction | repeated fast-action failure triggers s...  |        |
| GPM: governed persistent memory (Xu, 2026)                 | 2026 | ext. memory   | –           | symbolic check (source binding, retract...  |        |
| MemGym (Xu et al., 2026c)                                  | 2026 | ext. memory   | –           | none                                        |        |
| EnvACE world rehearsal (Xu et al., 2026e)                  | 2026 | world model   | trajectory  | task-success RL reward jointly optimize...  |        |
| META (Xu et al., 2026b)                                    | 2026 | ext. memory   | –           | trade outcome + LLM reflection; prune/c...  |        |
| Representation signatures in trad... (Xue, 2026)           | 2026 | context       | output      | structured risk feedback                    |        |
| MemStrata temporal validity (Yadav, 2026)                  | 2026 | ext. memory   | –           | contradiction of a stored (subject, rela... | ✓      |
| Memory-R2 / LoGo-GRPO (Yan et al., 2026a)                  | 2026 | ext. memory   | state       | local rerollouts from shared intermedia...  |        |
| DenoiseFlow (Yan et al., 2026b)                            | 2026 | context       | trajectory  | semantic entropy over samples               |        |
| Belief Engine stance dynamics (Yang et al., 2026c)         | 2026 | prob. memory  | state       | log-odds update with evidence uptake an...  | ✓      |
| GroupMemBench multi-party memory (Yang et al., 2026b)      | 2026 | ext. memory   | –           | none                                        |        |
| Graph-based agent memory survey (Yang et al., 2026a)       | 2026 | ext. memory   | –           | memory evolution stage (update/consolid...  |        |
| EvoTool (Yang et al., 2026e)                               | 2026 | context       | interaction | none                                        |        |
| world models as intermediary (Yang, 2026)                  | 2026 | world model   | WM loss     | none (position paper)                       |        |
| EnterpriseBench (Yang et al., 2026d)                       | 2026 | context       | –           | none (benchmark)                            |        |
| Werewolf accusation belief-shift (Yang et al., 2026f)      | 2026 | written       | –           | LLM in-context update after accusation...   | ✓      |
| Forecast-Dojo (Ye et al., 2026)                            | 2026 | written       | trajectory  | LM rewrites JSON belief notebook (proba...  | ✓      |
| AgeMem (Yu et al., 2026b)                                  | 2026 | ext. memory   | memory op   | agent-initiated store/retrieve/update/s...  | ○      |
| LOCARD structured belief state (Yu and Knottenbelt, 2026)  | 2026 | written       | –           | strategic core flips a boolean SOP-mile...  |        |
| InfAgent (Yu et al., 2026a)                                | 2026 | written       | –           | none                                        |        |
| SAVeR: self-audited verified reas... (Yuan et al., 2026a)  | 2026 | context       | state       | adversarial audit locating logical/evid...  |        |
| RPMS (Yuan et al., 2026b)                                  | 2026 | written       | –           | lightweight belief state gates memory a...  | ○      |
| Critic Experience Bank (Zeng et al., 2026)                 | 2026 | ext. memory   | –           | hindsight LLM majority vote conditioned...  | ✓      |
| Credit Assignment in RL for LLMs... (Zhang, 2026)          | 2026 | –             | trajectory  | n/a                                         |        |
| MICA: intertemporal credit for em... (Zhang et al., 2026d) | 2026 | ext. memory   | interaction | none (potential over structured user st...  |        |
| Q-Evolve (Zhang et al., 2026k)                             | 2026 | context       | trajectory  | advantage from in-distribution IQL crit...  |        |
| Internalizing the Future (Zhang et al., 2026h)             | 2026 | written       | trajectory  | foresight-conditioned RL calibrating pl...  |        |
| RetailBench long-horizon retail P... (Zhang et al., 2026c) | 2026 | context       | –           | none                                        |        |
| WorldEvolver (Zhang et al., 2026g)                         | 2026 | world model   | state       | prediction-observation mismatch             |        |
| Theory of Space (Zhang et al., 2026f)                      | 2026 | written       | –           | new partial observations during active...   | ✓      |
| MemSkill (Zhang et al., 2026b)                             | 2026 | ext. memory   | memory op   | controller selects extract/consolidate/...  | ○      |
| OASES (Zhang et al., 2026a)                                | 2026 | context       | state       | none                                        |        |
| HiMem (Zhang et al., 2026e)                                | 2026 | ext. memory   | –           | conflict detected between retrieved not...  | ○      |
| StarWM (Zhang et al., 2026i)                               | 2026 | world model   | WM loss     | offline next-observation prediction fro...  | ○      |
| CORE (+PERSIST) (Zhang et al., 2026j)                      | 2026 | prob. memory  | state       | known answer (gold slot values: state-f...  | ✓      |
| AEM adaptive entropy modulation (Zhao et al., 2026a)       | 2026 | context       | trajectory  | response-level surprisal x advantage        |        |
| EvoSCM causal belief revision (Zhao et al., 2026b)         | 2026 | written       | state       | prediction-observation discrepancy dist...  | ✓      |
| BranPO (Zhao et al., 2026c)                                | 2026 | –             | interaction | none                                        |        |
| PGPO (Zheng et al., 2026c)                                 | 2026 | context       | interaction | potential difference between consecutiv...  |        |
| FOREAGENT predict-before-execute (Zheng et al., 2026a)     | 2026 | world model   | –           | predict-then-verify: prediction replace...  | ○      |
| Code2World (Zheng et al., 2026b)                           | 2026 | world model   | WM loss     | render-aware RL rewards rendered next-s...  | ○      |
| Thinking-RFT for ToM (Zhong et al., 2026a)                 | 2026 | –             | output      | verifiable-reward RL on shortcut-free T...  | ○      |
| LSE-MTP (Zhong et al., 2026b)                              | 2026 | context       | WM loss     | multi-token prediction anchored to grou...  | ○      |
| ToolVerse (Zhou et al., 2026)                              | 2026 | context       | interaction | turn-aware relative advantage from outc...  |        |
| Simple mechanism interfaces (Zhu et al., 2026)             | 2026 | context       | –           | none                                        |        |
| Qwen-AgentWorld (Zuo et al., 2026)                         | 2026 | world model   | WM loss     | next-state prediction reasoning trained...  | ✓      |

Metric: ✓ belief-level metric reported, ○ proxy only.



**Fig. 6 | Where credit has gone, 2020–2026.** Lanes are credited objects; Table 15 lists every screened system. Written beliefs exist from dialogue state tracking onward, but until 2026 they are credited by task outcome or not trained at all; the red points are the first to give the belief its own learning signal.